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Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$

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Abstract
Let $p \geq 3$ be a prime number. In this article, we study the canonical expansion of the primitive p^n-th root of unity ζ_{p^n} in p-adic Mal'cev–Neumann field $\mathbb{L}_p$ for $n \geq 1$. More precisely, we give the explicit formula for the first $\aleph_0$ terms of the expansion of ζ_{p^n}, and as an application, we use it to construct a uniformizer of $K_{2,m} = \mathbb{Q}_p\left(\zeta_{p^2}, p^{1/p^m}\right)$ with $m \geq 1$.

Keywords False Tate extension · Explicit uniformizer · p-Adic Mal'cev–Neumann field · Newton algorithm · Newton polygon

Mathematics Subject Classification 11B73 · 11C08 · 11D88 · 11P83 · 11S20 · 11T22 · 11Y40 · 12E30 · 12F10 · 41A58

1 Introduction

1.1 Motivation

Let $p \geq 3$ be a prime number. For an integer $n \geq 1$, let μ_{p^n} be the group of p^n-th roots of unity. We fix a compatible system $\epsilon = (\zeta_{p^n} \in \mu_{p^n})_{n \geq 0}$ of primitive p^n-th root of unity (i.e. for any $l \leq n$, we have $\zeta_{p^n}^{p^l} = \zeta_{p^{n-l}}$). For $n \geq m \geq 0$ two integers, we denote by $K_{n,m} = \mathbb{Q}_p\left(\mu_{p^n}, p^{1/p^m}\right)$ the false Tate curve extension of $\mathbb{Q}_p$, which

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is a finite Galois extension of $\mathbb{Q}_p$ of degree $\varphi(p^n)p^m$. Let Γ be the Galois group of $\mathbb{Q}_p^{\mathrm{cycl}} = \cup_n K_{n,0}$ over $\mathbb{Q}_p$ and let Γ^{FT} be the Galois group of $K_\infty = \cup_n K_{n,n}$ over $\mathbb{Q}_p$. Both of them are p-adic Lie groups.

Let $\mathcal{O}_{\mathcal{E}}^+ = \mathbb{Z}_p[[T]]$, $\mathcal{O}_{\mathcal{E}}$ the p-adic completion of $\mathcal{O}_{\mathcal{E}}^+[\frac{1}{T}]$ and $\mathcal{E} = \mathcal{O}_{\mathcal{E}}^+[\frac{1}{p}]$ the fraction field of $\mathcal{O}_{\mathcal{E}}$. The field $\mathcal{E}$ is equipped with actions of φ and Γ given by the formulae:

$$\varphi(T) = (1+T)^p - 1, \gamma(T) = (1+T)^{\chi_{\mathrm{cycl}}(\gamma)},$$

where χ_{cycl} is the cyclotomic character. An étale (φ, Γ)-module over $\mathcal{E}$ is a finite dimensional $\mathcal{E}$-vector space D endowed with semi-linear actions of φ and Γ commuting with each other, such that $\varphi^*D \cong D$. If D is an étale (φ, Γ)-module over $\mathcal{E}$, all the elements x of D can be uniquely written in the form $x = \sum_{i=0}^{p-1} \varphi(x_i)(1+T)^i$, with $x_i \in \mathcal{E}$. This allows us to define a left inverse $\psi : D \to D$ of φ by the formula $\psi(x) = x_0$; moreover, ψ commutes with Γ.

The theory of (φ, Γ)-modules introduced by Fontaine is fruitful for the study of commutative Iwasawa theory. Among the other things, the fundamental result is the following theorem (cf. [3, Théorème II.1.3]):

Theorem 1.1 (Fontaine) *For any p-adic representation V, there is an isomorphism of $\mathbb{Z}_p[[\Gamma]]$-modules*

$$\mathrm{Exp}^* : \mathrm{H}^1_{\mathrm{Iw}}(\mathbb{Q}_p, V) \cong D(V)^{\psi=1},$$

where $\mathrm{H}^1_{\mathrm{Iw}}(\mathbb{Q}_p, V) = \mathrm{H}^1\left(\mathrm{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p), \mathbb{Z}_p[[\Gamma]] \otimes V\right)$ *and $D(V)$ is the (φ, Γ)-module associated to V by Fontaine's equivalence of categories* (cf. [7, Théorème 3.4.3]).

In 2004, Coates et al. [4] proposed a programme of non-commutative Iwasawa theory. In view of the important role played by the theory of (φ, Γ)-modules in commutative Iwasawa theory, it is natural to ask if there is an analogy of the (φ, Γ)-module theory in the non-commutative situation. The first interesting case can be the tower of the false Tate curve extension of $\mathbb{Q}_p$. In [16], Ribeiro introduced the notion of cohomology of $(\varphi, \Gamma^{\mathrm{FT}})$-modules. But this definition seems very difficult to describe non-commutative Iwasawa cohomology. A more direct way could be intimating the theory of field of norms of Fontaine and Wintenberger in this case and rebuild the whole theory. One surprising obstruction is that we do not even know how to write down a norm-compatible system of uniformizers of the tower $\{K_{n,m}\}_{n\geq m\geq 0}$ explicitly, which is a key input in the theory of field of norms.

Recently there have been some attempts to attack this problem. In [17], Viviani gave a uniformizer of $K_{1,m}$:

$$\pi_{1,m} = \frac{1-\zeta_p}{\prod_{i=1}^m p^{\frac{1}{p^i}}}.$$

If we denote by $v_{1,m}$ the p-adic valuation on $K_{1,m}$ normalized by $v_{1,m}(p) = p^m(p-1)$, then $v_{1,m}(1-\zeta_p) = p^m$ and $v_{1,m}(p^{\frac{1}{p^m}}) = p-1$ which are coprime to each other. Thus, one can use Bézout's lemma to construct a uniformizer in this case. Bellemare and Lei [1] expand an idea of the user "Mercio" on the website Stackexchange and constructed a uniformizer for the field $K_{2,1}$, and they explain the reason why their method cannot go further. In this article, we extend an idea of Lampert (cf. [13]) to construct a uniformizer of $K_{2,m}$ with $m \geq 1$.

1.2 Main results

Convention Let $[\cdot] : \bar{\mathbb{F}}_p \to W(\bar{\mathbb{F}}_p) = \mathcal{O}_{\breve{\mathbb{Q}}_p}$ be the Teichmüller character, where $W(\bar{\mathbb{F}}_p)$ is the ring of Witt vectors over $\bar{\mathbb{F}}_p$. For any positive integer k that is coprime to p, by abuse of notation, we will not distinguish the symbol of kth primitive root ζ_k in $\bar{\mathbb{F}}_p$ and its Teichmüller representative $[\zeta_k]$ in $\mathcal{O}_{\breve{\mathbb{Q}}_p}$.

As we observed in the case $K_{1,m}$, if one can find an algebraic integer of $K_{n,m}$ with valuation coprime to p, then we can use Bézout's lemma to construct a uniformizer of $K_{n,m}$.

David Lampert in his paper [11] gave the p-adic expansion of ζ_{p^2} without a proof.[1] The formula appearing in his paper indicates (cf. [13]) that there is a chance to construct the desired algebraic integer. This leads us to study the canonical expansion of the primitive root of unity ζ_{p^n} in the p-adic Mal'cev–Neumann field $\mathbb{L}_p$, which is the spherical completion of $\mathbb{C}_p$ (cf. Sect. 2.1). On the other hand, Kedlaya [9] used a transfinite induction to prove the algebraic closeness of the p-adic Mal'cev–Neumann field $\mathbb{L}_p$. We expand Kedlaya's proof into a transfinite Newton's algorithm in Sect. 2. We will call the result of the ith step of the transfinite Newton algorithm for a given polynomial $P(T) \in \mathbb{L}_p[T]$, the ith approximation of a root of $P(T)$. Using this algorithm, we prove an explicit formula for the first $\aleph_0$ terms of canonical expansion of a p^n-th primitive root of unity in $\mathbb{L}_p$ for every $n \geq 2$ (cf. Theorem 3.3 of local cite):

Theorem *Let $\zeta_{p^n}^{(i)}$ in $\mathbb{L}_p$ be the ith approximation of ζ_{p^n} in the transfinite Newton algorithm for all $n \geq 2$. Then, we have*

$$\zeta_{p^n}^{(i)} = \begin{cases} \sum_{k=0}^{i} \frac{(-1)^{kn}}{[k!]} \zeta_{2(p-1)}^{k} p^{\frac{k}{p^{n-1}(p-1)}} & \text{for } 0 \leq i \leq p-1, \\ \zeta_{p^n}^{(p-1)} + \sum_{l=n}^{i-p+n} (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^l}} & \text{for } i \geq p. \end{cases}$$

In other words, we have

$$\zeta_{p^n} = \sum_{i=0}^{p-1} \frac{(-1)^{in}}{[i!]} \zeta_{2(p-1)}^{i} p^{\frac{i}{p^{n-1}(p-1)}} + \sum_{i=n}^{\infty} (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^i}} + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right).$$

[1] Lampert claimed at [12,13] that the expansion in his paper is incorrect.

In the same theorem (cf. Theorem 3.3), we discuss the relation among all the possibilities of the first $\aleph_0$ terms of p^n-th primitive roots:

Theorem *For every $\alpha \in \mathbb{L}_p$, denote by $\aleph_0(\alpha)$ the first $\aleph_0$ terms of the expansion of α in $\mathbb{L}_p$.*

1. *There exists $p-1$ distinct elements $m_0, \ldots, m_{p-2}$ in $(\mathbb{Z}/p^n\mathbb{Z})^\times$ such that*

   (a)

$$\aleph_0\left(\zeta_{p^n}^{m_k}\right) = \sum_{i=0}^{p-1} \frac{\left((-1)^n \zeta_{2(p-1)}^{2k+1}\right)^i}{[i!]} p^{\frac{i}{p^{n-1}(p-1)}} + \sum_{j=n}^{\infty} (-1)^n \zeta_{2(p-1)}^{2k+1} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^j}};$$

   (b) $\mathcal{R}_n = \{m_0, \ldots, m_{p-2}\}$ *forms a* mod p *residue system of* $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

2. *For every $m \in (\mathbb{Z}/p^n\mathbb{Z})^\times$, there exists a unique $m_t \in \mathcal{R}_n$ such that $\aleph_0\left(\zeta_{p^n}^{m}\right) = \aleph_0\left(\zeta_{p^n}^{m_t}\right)$.*

Besides that, we give an analogous result for ζ_p in Proposition 3.4. Finally, using Theorem 3.3, we construct a uniformizer of $K_{2,m}$ (cf. Theorem 3.23 of local cite):

Theorem 1. *The element*

$$\pi_{2,1} = \left(p^{\frac{1}{p}}\right)^{-1} \left(\zeta_{p^2} - \sum_{k=0}^{p-1} \frac{1}{[k!]} \zeta_{2(p-1)}^{k} p^{\frac{k}{p(p-1)}}\right)$$

is a uniformizer of $K_{2,1}$.

2. *For $m \geq 2$, the element*

$$\pi_{2,m} = \left(p^{\frac{1}{p^m}}\right)^{-\frac{p^m-1}{p-1}} \left(\zeta_{p^2} - \sum_{k=0}^{p-1} \frac{1}{[k!]} \zeta_{2(p-1)}^{k} p^{\frac{k}{p(p-1)}} - \sum_{l=2}^{m} \zeta_{2(p-1)} p^{\frac{1}{p-1} - \frac{1}{p^l}}\right)$$

is a uniformizer of $K_{2,m}$.

Remark 1.2 If one can give the explicit formula for the second $\aleph_0$ terms of the canonical expansion of the p^n-th primitive root of unity in $\mathbb{L}_p$, then it is possible that our strategy can go further to find a uniformizer in more general cases.

2 Transfinite Newton algorithm

In this paragraph, we summarize the properties of the p-adic Mal'cev–Neumann field $\mathbb{L}_p$ in Sect. 2.1, and expand Kedlaya's proof of the algebraic closeness of $\mathbb{L}_p$ into a transfinite Newton algorithm in Sect. 2.2.

2.1 The p-adic Mal'cev–Neumann field $\mathbb{L}_p$

Let $\mathcal{O}_{\breve{\mathbb{Q}}_p} = W(\bar{\mathbb{F}}_p)$ be the ring of Witt vectors over $\bar{\mathbb{F}}_p$ and let $\mathbb{L}_p$ be the p-adic Mal'cev–Neumann field $\mathcal{O}_{\breve{\mathbb{Q}}_p}((p^{\mathbb{Q}}))$ (cf. [15, Section 4]). Every element α of $\mathbb{L}_p$ can be uniquely written as follows:

$$\sum_{x\in\mathbb{Q}}[\alpha_x]p^x,\ \text{where } [\cdot]:\bar{\mathbb{F}}_p \to W(\bar{\mathbb{F}}_p) \text{ is the Teichmüller character.} \tag{2.1}$$

For any $\alpha = \sum_{x\in\mathbb{Q}}[\alpha_x]p^x \in \mathbb{L}_p$, we set $\mathrm{Supp}(\alpha) = \{x \in \mathbb{Q} : \alpha_x \neq 0\}$, which is well-ordered by the definition of $\mathbb{L}_p$. Thus, we can define the p-adic valuation v_p by the formula:

$$v_p(\alpha) = \begin{cases} \inf \mathrm{Supp}(\alpha) & \text{if } \alpha \neq 0; \\ \infty & \text{if } \alpha = 0 \end{cases}.$$

The field $\mathbb{L}_p$ is complete for the p-adic topology, and it is also algebraically closed. Moreover, it is the maximal complete immediate extension[2] of $\overline{\mathbb{Q}}_p$.

Remark 2.1 $\mathbb{L}_p$ is spherical complete,[3] and $\mathbb{C}_p$ is not spherical complete. The field $\mathbb{C}_p$ of p-adic complex numbers can be continuously embedded into $\mathbb{L}_p$.

Given $\alpha \in \mathbb{L}_p$, for $x \in \mathbb{Q}$, we denote the coefficient of p^x in the expansion of α by $[C_x(\alpha)] \in \mathcal{O}_{\breve{\mathbb{Q}}_p}$. Then $C_x(\alpha) \in \bar{\mathbb{F}}_p$ is equal to $[C_x(\alpha)]$ modulo p. This gives a map

$$C : \mathbb{Q} \times \mathbb{L}_p \to \bar{\mathbb{F}}_p;\ (x, \alpha) \mapsto C_x(\alpha).$$

The following lemma summarizes the basic properties of the map C.

Lemma 2.2 *For every $x, y \in \mathbb{Q}$ and $\alpha, \beta \in \mathbb{L}_p$, we have*

1. *if $v_p(\alpha) > x$, then $C_x(\alpha) = 0$;*
2. *$C_x(p^{-y}\alpha) = C_{x+y}(\alpha)$;*
3. *for every $\bar{u} \in \bar{\mathbb{F}}_p$ and $u = [\bar{u}] \in \mathcal{O}_{\breve{\mathbb{Q}}_p}$, we have $\bar{u}C_x(\alpha) = C_x(u\alpha)$;*
4. *if $v_p(\alpha), v_p(\beta) \geq x$, then $C_x(\alpha) \pm C_x(\beta) = C_x(\alpha \pm \beta)$.*

2.2 Transfinite Newton algorithm

Definition 2.3 (Newton polygon) Suppose (K, v) is a valued field with value group $\mathbb{Q}$. Let $J(T) = \sum_{i=0}^{n} a_{n-i}T^i \in K[T]$ be a nonzero polynomial. For $0 \leq i \leq n$, we have the points $(i, v(a_i)) \in \mathbb{N} \times \bar{\mathbb{R}}$, where $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$. If $a_i = 0$, $(i, v(a_i))$ is regarded as $Y_{+\infty}$, the point at infinity of the positive vertical axis.

[2] A valued field extension (E, w) of (F, v) is an immediate extension, if (E, w) and (F, v) have the same residue field. A valued field (E, w) is maximally complete if it has no immediate extensions other than (F, v) itself.

[3] A valued field is said to be spherical complete, if the intersection of every decreasing sequence of closed balls is nonempty.

1. Define the **Newton polygon** $\mathscr{N}ewt\,(J)$ of $J(T)$ as the lower boundary of the convex hull of the points $(i, v(a_i))$ for $i = 0, \ldots, n$. As a consequence, $\mathscr{N}ewt\,(J)$ is a function on $\mathbb{R}_{\geq 0}$ with values in $\bar{\mathbb{R}}$.
2. The integers m such that $(m, v(a_m))$ are vertices of $\mathscr{N}ewt\,(J)$ are called the **breakpoints**, and we denote by $m^J_{\max}$ the **largest breakpoint** less than n.
3. Given two adjacent breakpoints $m^J_1 < m^J_2$, denote by $s^J_{m_1} = (v(a_{m^J_2}) - v(a_{m^J_1}))/(m^J_2 - m^J_1)$, the **slope** of constituent segment of $\mathscr{N}ewt\,(J)$ with endpoints $(m^J_1, v(a_{m^J_1}))$ and $(m^J_2, v(a_{m^J_2}))$. The **largest slope** is denoted by $s^J_{\max} = s^J_{m^J_{\max}} = (v(a_n) - v(a_{m^J_{\max}}))/(n - m^J_{\max})$. If $(n, v(a_n)) = Y_{+\infty}$ (i.e. $a_n = 0$),[4] we regard $s^J_{\max} = \infty$. Thus, $s^{\bullet}_{\max}$ is a map from $K[T]$ to $\mathbb{Q} \cup \{\infty\}$.

We will omit the superscript J if there is no confusion.

Let $P(T) = a_0T^n + a_1T^{n-1} + \cdots + a_n \in \mathbb{L}_p[T]$ be a polynomial with $a_n \neq 0$. For any $u \in \mathcal{O}^*_{\mathbb{L}_p}$, set

$$P_u(T) = P(T + up^{s^P_{\max}}),$$

where $s^P_{\max}$ is the maximal slope of the Newton polygon of P.

Lemma 2.4 *Let $P(T) = a_0T^n + a_1T^{n-1} + \cdots + a_n \in \mathbb{L}_p[T]$ be a polynomial with $a_n \neq 0$. For any $u \in \mathcal{O}^*_{\mathbb{L}_p}$, we write $P_u(T) = \sum_{i=0}^n b_{n-i}T^i$.*

Then one has

1. *The Newton polygons $\mathscr{N}ewt\,(P_u)$ and $\mathscr{N}ewt\,(P)$ are identical in the range $\left[0, m^P_{\max}\right]$;*
2. *If $m^P_{\max} < k \leq n$, then the point $(k, v_p(b_k))$ is on or above $\mathscr{N}ewt\,(P)$, in other words, we have*

$$v_p(b_k) \geq v_p(a_{m^P_{\max}}) + s^P_{\max}(k - m^P_{\max}).$$

Proof For simplification of notations, we set $s = s^P_{\max}$ and $m = m^P_{\max}$. We calculate the p-adic valuation of

$$b_k = \sum_{j=0}^{k} a_{k-j}\binom{n-k+j}{j}u^j p^{sj}.$$

Note that $v_p\left(a_{k-j}\binom{n-k+j}{j}u^j p^{sj}\right) = v_p(a_{k-j}) + sj$.

1. Suppose $k \leq m$ is a breakpoint of $\mathscr{N}ewt\,(P)$. If $j > 0$, one observes

$$v_p(a_{k-j}) + sj = v_p(a_k) + j\left(s - \frac{v_p(a_k) - v_p(a_{k-j})}{k - (k-j)}\right).$$

[4] Notice that if m is a breakpoint, then $(m, v(a_m)) = Y_{+\infty} \Leftrightarrow m = n$ and $a_n = 0$.

Since s is the maximal slope of $\mathscr{N}ewt\,(P)$ and i is a breakpoint, one has $(v_p(a_k) - v_p(a_{k-j}))/(k-(k-j)) < s$. In other words, for all $j > 0$, we have $v_p\left(a_{k-j}\binom{n-k+j}{j}u^j p^{sj}\right) > v_p(a_k)$. As a consequence, in this case, we have $v_p(b_k) = v_p(a_k)$.

Now suppose that $k < m$ is not a breakpoint of $\mathscr{N}ewt\,(P)$. Let $m_1^P < m_2^P$ be two adjacent breakpoints of P such that $m_1^P < k < m_2^P$. We claim that for all $0 \le j \le k$, we have

$$v_p(a_{k-j}) + sj \ge (k - m_1^P)s_{m_1^P} + v_p(a_{m_1^P}). \tag{2.2}$$

This claim implies that

$$v_p(b_k) \ge (k - m_1^P)s_{m_1^P} + v_p(a_{m_1^P}),$$

i.e. the point $(k, v_p(b_k))$ is on or above $\mathscr{N}ewt\,(P)$.
In the following, we prove the claim (2.2). Since $s_{m_1^P} < s$, one has

$$sj - (k - m_1^P)s_{m_1^P} \ge s_{m_1^P}(m_1^P - (k-j)).$$

(a) If $k - j = m_1^P$, we have

$$\begin{aligned} v_p(a_{k-j}) + sj &= v_p(a_{m_1^P}) + sj \\ &\ge v_p(a_{m_1^P}) + s_{m_1^P} j \\ &= v_p(a_{m_1^P}) + s_{m_1^P}(k - m_1^P). \end{aligned}$$

(b) If $k - j < m_1^P$, we have

$$\frac{sj - (k - m_1^P)s_{m_1^P}}{m_1^P - (k-j)} \ge s_{m_1^P} \ge \frac{v_p(a_{m_1^P}) - v_p(a_{k-j})}{m_1^P - (k-j)}.$$

(c) If $k - j > m_1^P$, one has

$$\frac{v_p(a_{k-j}) - v_p(a_{m_1^P})}{(k-j) - m_1^P} \ge s_{m_1^P} \ge \frac{s_{m_1^P}(k - m_1^P) - sj}{(k-j) - m_1^P}.$$

2. The second assertion follows from the same discussion.

□

Definition 2.5 For any polynomial $P(T) = a_0T^n + a_1T^{n-1} + \cdots + a_n \in \mathbb{L}_p[T]$, we set $s = s_{\max}^P$ and $m = m_{\max}^P$. We define a polynomial

$$\mathrm{Res}_P(T) = \sum_{k=0}^{n-m} C_0\left(a_{n-k}p^{-v_p(a_m)-s(n-m-k)}\right)T^k \in \bar{\mathbb{F}}_p[T],$$

called the *residue polynomial* associated to $P(T)$.

Remark 2.6 From the geometric point of view, the residue polynomial can be constructed as follows:

1. Intercept the segment with maximal slope.
2. Record those coefficients a_i of $P(T)$ with $(i, v_p(a_i))$ lying on this segment as b_i. The other coefficients b_i should be recorded as 0.
3. Using the coefficients (b_i) in the previous step and the map $\mathbb{L}_p \to \bar{\mathbb{F}}_p, \alpha \mapsto C_{v_p(\alpha)}(\alpha)$, one can construct the polynomial

$$\mathrm{Res}_P(T) = \sum_{k=0}^{n-m} C_{v_p(a_{n-k})}(a_{n-k})\, T^k \in \bar{\mathbb{F}}_p[T].$$

Note that $v_p(a_{n-k}) = v_p(a_m) + s(n-m-k)$, and we have $C_{v_p(\alpha)}(\alpha) = C_0(\alpha \cdot p^{-v_p(\alpha)})$ by Lemma 2.2.

Proposition 2.7 *Let* $P(T) = a_0T^n + a_1T^{n-1} + \cdots + a_n \in \mathbb{L}_p[T]$ *be a polynomial with* $a_n \neq 0$ *and* $\mathrm{Res}_P(T) \in \bar{\mathbb{F}}_p[T]$ *its residue polynomial. Let* $c \in \bar{\mathbb{F}}_p$ *be a root of* $\mathrm{Res}_P(T)$ *with multiplicity* q. *We set*

$$P_{[c]}(T) = P(T + [c]p^{s_{\max}^P}) = \sum_{i=0}^{n} b_{n-i}T^i.$$

Then, we have

1. $n - q$ *is a breakpoint of* $\mathcal{N}ewt\left(P_{[c]}\right)$*;*
2. *in the range* $[0, n-q]$, $\mathcal{N}ewt\,(P)$ *is identical with* $\mathcal{N}ewt\left(P_{[c]}\right)$*;*
3. *the remaining slope(s) of* $\mathcal{N}ewt\left(P_{[c]}\right)$ *are strictly greater than* $s_{\max}^P$.

Proof Set $s = s_{\max}^P$ and $m = m_{\max}^P$. Recall that, since m is the maximal breakpoint of the Newton polygon $\mathcal{N}ewt\,(P)$, for $m \leq n-k$, we have

$$v_p(a_{n-k}p^{-v_p(a_m)-s(n-m-k)}) = v_p(a_{n-k}) - v_p(a_m) - s(n-m-k) \geq 0.$$

Thus, $C_0\left(a_{n-k}p^{-v_t(a_m)-s(n-m-k)}\right) \in \bar{\mathbb{F}}_p$ is the image of $a_{n-k}p^{-v_p(a_m)-s(n-m-k)}$ under the canonical projection from $\mathbb{L}_p$ to its residue field $\bar{\mathbb{F}}_p$.

Let $\mathrm{Res}_P(T+c) = \sum_{k=0}^{n-m} \kappa_{n-k}T^k$, then we have

$$\begin{aligned}\kappa_{n-k} &= \sum_{i=n-k}^{n-m} C_0\left(a_{n-i}p^{-v_p(a_m)-s(n-m-i)}\right)\binom{i}{n-k}c^{i-(n-k)} \\ &= \sum_{j=0}^{k-m} C_0\left(p^{-v_p(a_m)-s(k-m)}a_{k-j}p^{sj}\right)\binom{n-k+j}{j}c^j.\end{aligned} \tag{2.3}$$

By the basic properties of the map $C_x(\alpha)$ (cf. Lemma 2.2), we have

$$
\begin{aligned}
2.3 &= \sum_{j=0}^{k-m} \binom{n-k+j}{j} c^j C_{v_p(a_m)+s(k-m)}\left(a_{k-j}p^{sj}\right) \\
&= \sum_{j=0}^{k-m} \binom{n-k+j}{j} C_{v_p(a_m)+s(k-m)}\left([c^j]a_{k-j}p^{sj}\right).
\end{aligned} \tag{2.4}
$$

A similar argument in the proof of (2.2) in Lemma 2.4 shows that

$$
v_p(a_{k-j}) + sj \begin{cases} > v_p(a_m) + s(k-m) & \text{if } j > k-m \\ \geq v_p(a_m) + s(k-m) & \text{if } j \leq k-m. \end{cases}
$$

Again by Lemma 2.2, for $j > k-m$, we have $C_{v_p(a_m)+s(k-m)}\left([c^j]a_{k-j}p^{sj}\right) = 0$, and

$$
\begin{aligned}
2.4 &= \sum_{j=0}^{k} \binom{n-k+j}{j} C_{v_p(a_m)+s(k-m)}\left([c^j]a_{k-j}t^{sj}\right) \\
&= C_{v_p(a_m)+s(k-m)}\left(\sum_{j=0}^{k} \binom{n-k+j}{j} c^j a_{k-j}t^{sj}\right) \\
&= C_{v_p(a_m)+s(k-m)}\left(b_k\right).
\end{aligned}
$$

Since $a_m = b_m$, one can conclude that, for $0 \leq k \leq n-m$, the coefficient κ_k of T^{n-k} in $\mathrm{Res}_P(T+c)$ is equal to $C_{v_p(b_m)+s(k-m)}(b_k)$. Since c, as a root of $\mathrm{Res}_P(T)$, has multiplicity q, T^q has nonzero coefficient in $\mathrm{Res}_P(T+c)$, i.e. we have

$$
\kappa_{n-q} = C_{v_p(b_m)+s(n-q-m)}(b_k)(b_{n-q}) \neq 0.
$$

On the other hand, we have $v_p(b_{n-q}) \geq v_p(b_m) + s(n-q-m)$. Thus, we have

$$
v_p(b_{n-q}) = v_p(b_m) + s(n-q-m).
$$

If $k > n-q$, the coefficient κ_k of T^{n-k} in $\mathrm{Res}_P(T+c)$ is 0; thus, $v_p(b_k) > v_t(b_m) + s(k-m)$.

□

Theorem 2.8 (Kedlaya) *The field $\mathbb{L}_p$ is algebraically closed.*

Proof Let $f(T)$ be a non-constant polynomial in $\mathbb{L}_p[T]$. To construct a root of $f(T)$ in $\mathbb{L}_p$ is equivalent to find an element $r \in \mathbb{L}_p$ such that the maximal slope of the Newton polygon $\mathscr{Newt}\,(f(T+r))$ is ∞.

Let $\aleph_1$ be the minimal uncountable ordinal. We define a sequence of elements $(r_w)_{w<\aleph_1}$ in $\mathbb{L}_p$ by transfinite induction.

- For $w=0$, we set $r_0=0$ and let $\tau(0)$ be a root of $\mathrm{Res}_f(T)$ in $\bar{\mathbb{F}}_p$ and $d(0)$ be the maximal slope of the Newton polygon $\mathscr{N}ewt\,(f(T+0))$.
- Let $0<\omega<\aleph_1$ be a successor ordinal. If $f(r_{\omega-1})=0$, then we set $r_\omega = r_{\omega-1} = r_{\omega-1}+[0]\cdot p^\infty$, i.e. $\tau(\omega-1)=0$ and $d(\omega-1)=\infty$. Now suppose $f(r_{\omega-1})\neq 0$ and r_α has been constructed for any ordinal $\alpha<\omega$. Define $\tau(\omega-1)$ to be a root of $\mathrm{Res}_{f(T+r_{\omega-1})}(T)$ in $\bar{\mathbb{F}}_p$ and $d(\omega-1)$ to be the maximal slope of $\mathscr{N}ewt\,(f(T+r_{\omega-1}))$. Then we set $r_\omega=r_{\omega-1}+[\tau(\omega-1)]p^{d(\omega-1)}\in\mathbb{L}_p$.
- For limit ordinal $\omega_{\lim}$, we set $r_{\omega_{\lim}}=\sum_{\alpha<\omega_{\lim}}[\tau(\omega)]p^{d(\alpha)}\in\mathbb{L}_p$.

For all ordinals $\alpha<\beta$, by Proposition 2.7, we know that $d(\alpha)\leq d(\beta)$. Moreover, we have $d(\alpha)=d(\beta)$ if and only if $d(\omega)=\infty$ holds for all $\omega\geq\alpha$. Since there is no injection from $\aleph_1$ to $\mathbb{Q}$, there exist an ordinal $\phi<\aleph_1$ such that $d(\phi)=\infty$, i.e. r_ϕ is a root of $f(T)$ in $\mathbb{L}_p$. □

We summarize the above theorem into the following algorithm[5]:

Algorithm 1 transfinite Newton algorithm for $\mathbb{L}_p$

INPUT: A non-constant polynomial $f(T)\in\mathbb{L}_p[T]$
OUTPUT: A root of $f(T)$ in $\mathbb{L}_p$
 function NEWTON(f)
  $r\leftarrow 0$
  $s_{\max}\leftarrow 0, m_{\max}\leftarrow 0, c\leftarrow 0$
  $\mathrm{Res}_\Phi(T)\leftarrow 0$
  $\Phi(T)\leftarrow f(T)$ ▷ We denote the coefficient of T^i in Φ as b_{n-i},where $n=\deg(\Phi)$.
  while $\Phi(0)\neq 0$ **do**
   $m_{\max}\leftarrow m^\Phi_{\max}$
   $s_{\max}\leftarrow s^\Phi_{\max}$
   $\mathrm{Res}_\Phi(T)\leftarrow\sum_{k=0}^{n-m_{\max}} C_{v_p(b_m)+s_{\max}(n-m_{\max}-k)}\left(b_{n-k}\right)T^k$
   $c\leftarrow$ any root of $\mathrm{Res}_\Phi(T)$ in $\bar{\mathbb{F}}_p$
   $r\leftarrow r+[c]\cdot p^{s_{\max}}$
   $\Phi(T)\leftarrow\Phi(T+[c]\cdot p^{s_{\max}})$
  end while
  return r
 end function

Definition 2.9 Let l be a natural number. We call the value of $\Phi(T)$ (resp. $\mathrm{Res}_\Phi(T)$ and r) in the above pseudocode after the loop iterates for l times, the lth approximation polynomial (resp. the lth residue polynomial and the lth approximation of a root of $\Phi(T)$).

[5] This piece of pseudocode dose not fit the definition of an algorithm in the sense of computer science, since it does not stop in finite steps and it does not treat the limit ordinal. We give this pseudocode here to clarify the main ingredient of Kedlaya's proof.

3 Application of the transfinite Newton algorithm to the p^n-th cyclotomic polynomial

Convention In this paragraph, we assume n is a positive integer.

Let $\Phi_{p^n}(T) = \sum_{k=0}^{p-1} T^{p^{n-1}k}$ be the p^n-th cyclotomic polynomial, whose Newton polygon is a segment of slope 0 with maximal breaking point $(0, 0)$. In Sect. 3.1, we apply the transfinite Newton algorithm on $\Phi_{p^n}(T)$ to get the first $\aleph_0$ terms of a p^n-th primitive root in $\mathbb{L}_p$ and discuss the relations among all the possibilities of the first $\aleph_0$ terms of p^n-th primitive roots for $n \geq 2$. Besides that, we give an analogous expansion for ζ_p. We establish our main combinatorial techniques in Sect. 3.2 and complete the proof of Theorem 3.3 in Sect. 3.3. Finally, using the expansion of ζ_{p^2}, we give a uniformizer of $K_{2,m}$ for $m \geq 2$ in Sect. 3.4.

3.1 First $\aleph_0$ terms of a root of $\Phi_{p^n}(T)$

The 0th residue polynomial of $\Phi_{p^n}(T)$ in the transfinite Newton algorithm is the polynomial $\mathfrak{A}_{0,n}(T) = \sum_{k=0}^{p-1} T^{p^{n-1}k}$, and the canonical element $1 \in \bar{\mathbb{F}}_p$ is a root of $\mathfrak{A}_{0,n}(T)$. With these initial inputs, the first approximation polynomial is

$$\Phi^{(1,n)}(T) = \sum_{k=0}^{p-1} (T+1)^{p^{n-1}k},$$

which has $p^{n-1}(p-1)$ roots with the same valuation $v_p(\zeta_{p^n} - 1) = \frac{1}{\varphi(p^n)} = \frac{1}{p^{n-1}(p-1)} > 0$. As a consequence, the Newton polygon $\mathscr{N}\!ewt\left(\Phi^{(1,n)}\right)$ (Fig. 1) is a segment of slope $\mathfrak{s}_{1,n} = \frac{1}{p^{n-1}(p-1)}$ with maximal breakpoint $\mathfrak{m}_{1,n} = (0, 0)$. Thus, the first residue polynomial is

$$\mathfrak{A}_{1,n}(T) = T^{p^{n-1}(p-1)} + 1 = (T^{p-1} + 1)^{p^{n-1}} \in \overline{\mathbb{F}}_p[T],$$

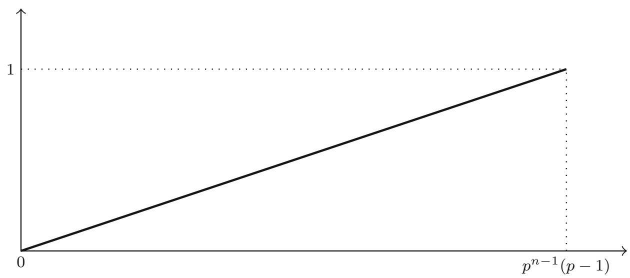

Fig. 1 $\mathscr{N}\!ewt\left(\Phi^{(1,n)}\right)$

which has $\mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)} \in \bar{\mathbb{F}}_p$ as a root[6] with multiplicity $\mathfrak{q}_{1,n} = p^{n-1}$.

Definition 3.1 The transfinite Newton algorithm can produce all the roots of a given polynomial $\Phi(T) \in \mathbb{L}_p[T]$. But note that we do not know whether the transfinite Newton algorithm can find all the roots of $\Phi_{p^n}(T)$, for $n \geq 2$, by repeatedly choosing different roots of the residue polynomial in each step. We can get at least $p-1$ roots of Φ_{p^n} in this way since there are $p-1$ choices of $\mathfrak{z}_{1,n}$ in $\bar{\mathbb{F}}_p$.

For the roots of $\Phi_{p^n}(T)$ constructed by the transfinite Newton algorithm, the second assertion of Theorem 3.3 below shows that their first $\aleph_0$ terms have exactly $p-1$ possibilities which are completely determined by the choice of $\mathfrak{z}_{1,n}$ in $\bar{\mathbb{F}}_p$. Moreover, for any root of $\Phi_{p^n}(T)$, there exists a root constructed by the algorithm such that they have the same first $\aleph_0$ terms.

Let ζ_{p^n} be a root of $\Phi_{p^n}(T)$ constructed by the transfinite Newton algorithm with input $\mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)}$. The following proposition summarizes the above discussion for the initial terms.

Proposition 3.2 *One has*

1. $\mathfrak{s}_{0,n} = 0 \in \mathbb{Q}$, $\mathfrak{z}_{0,n} = 1 \in \bar{\mathbb{F}}_p$;
2. $\mathfrak{s}_{1,n} = \frac{1}{p^{n-1}(p-1)}$, $\mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)} \in \bar{\mathbb{F}}_p$ *and the multiplicity of* $\mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)}$ *in* $\mathfrak{A}_1(T)$ *is* $\mathfrak{q}_{1,n} = p^{n-1}$.

In conclusion, the first approximation of ζ_{p^n} *is* $\zeta_{p^n}^{(1)} = \Lambda_{1,n}$, *where* $\Lambda_{1,n}$ *is defined to be* $1 + (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-1}(p-1)}}$. *In other words, we have*

$$\zeta_{p^n} = 1 + (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-1}(p-1)}} + o\left(p^{\frac{1}{p^{n-1}(p-1)}}\right).$$

For every $\alpha \in \mathbb{L}_p$, denote by $\aleph_0(\alpha)$ the first $\aleph_0$ terms of the expansion of α in $\mathbb{L}_p$. The following theorem gives the explicit formula for $\aleph_0\left(\zeta_{p^n}\right)$ with $n \geq 2$:

Theorem 3.3 *Let $n \geq 2$ be an integer.*

1. *Let* $\zeta_{p^n}^{(i)}$ *be the* i*th approximation of* ζ_{p^n} *in the transfinite Newton algorithm. Then we have*

$$\zeta_{p^n}^{(i)} = \begin{cases} \sum_{k=0}^{i} \frac{(-1)^{kn}}{[k!]} \zeta_{2(p-1)}^k p^{\frac{k}{p^{n-1}(p-1)}} & \text{for } 0 \leq i \leq p-1, \\ \zeta_{p^n}^{(p-1)} + \sum_{l=n}^{i-p+n} (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^l}} & \text{for } i \geq p, \end{cases}$$

i.e.

$$\aleph_0\left(\zeta_{p^n}\right) = \sum_{i=0}^{p-1} \frac{\left((-1)^n \zeta_{2(p-1)}\right)^i}{[i!]} p^{\frac{i}{p^{n-1}(p-1)}} + \sum_{j=n}^{\infty} (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^j}}.$$

2. *There exist* $p-1$ *distinct elements* $m_0, \ldots, m_{p-2}$ *in* $(\mathbb{Z}/p^n\mathbb{Z})^\times$ *such that*

[6] Adding the sign $(-1)^n$ to the root gives us a chance to get a compatible system $\left(\zeta_{p^n}\right)_{n\geq 1}$ of p^n-th primitive roots, i.e. $\zeta_{p^n} = \zeta_{p^{n+1}}^p$ (cf. Corollary 3.6).

(a)
$$\aleph_0\left(\zeta_{p^n}^{m_k}\right)=\sum_{i=0}^{p-1}\frac{\left((-1)^n\zeta_{2(p-1)}^{2k+1}\right)^i}{[i!]}p^{\frac{i}{p^{n-1}(p-1)}}+\sum_{j=n}^{\infty}(-1)^n\zeta_{2(p-1)}^{2k+1}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^j}};$$

(b) $\mathcal{R}_n=\{m_0,\ldots,m_{p-2}\}$ *forms a* mod p *residue system of* $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

3. *For every* $m\in(\mathbb{Z}/p^n\mathbb{Z})^\times$, *there exists a unique* $m_t\in\mathcal{R}_n$ *such that* $\aleph_0\left(\zeta_{p^n}^{m}\right)=\aleph_0\left(\zeta_{p^n}^{m_t}\right)$.

Proof of 1: We sketch the proof of the first assertion and leave the technical details of each step in next sections. We denote the formula on the right-hand side of the theorem by $\Lambda_{i,n}$ and the ith approximation polynomial of ζ_{p^n} by

$$\Phi^{(i,n)}(T)=\Phi_{p^n}\left(T+\zeta_{p^n}^{(i-1)}\right)=\sum_{k=0}^{p^{n-1}(p-1)}b_{p^{n-1}(p-1)-k}^{(i,n)}T^k\in\mathbb{L}_p[T].\qquad(3.1)$$

Moreover, we denote by $\mathfrak{A}_{i,n}(T)\in\bar{\mathbb{F}}_p[T]$ the residue polynomial of $\Phi^{(i,n)}(T)$. By the transfinite Newton algorithm, it is crucial to determine the following data:

(a) The maximal slope $\mathfrak{s}_{i,n}$ of the Newton polygon of $\Phi^{(i,n)}(T)$, which gives the support of the desired expansion,
(b) The residue polynomial $\mathfrak{A}_{i,n}(T)$, whose root $\mathfrak{z}_{i,n}\in\bar{\mathbb{F}}_p$ with multiplicity $\mathfrak{q}_{i,n}$ gives the coefficient $\alpha_{\mathfrak{s}_{i,n}}$ of the desired expansion.

To prove the theorem, one only needs to check that the supports and the coefficients in the ith step do coincide with those of $\Lambda_{i,n}$. The strategy of the proof is the following:

(a) **Describe the initial terms** (cf. Proposition 3.2): In fact, we have $\mathfrak{s}_{0,n}=0$, $\mathfrak{s}_{1,n}=\frac{1}{p^{n-1}(p-1)}$, $\mathfrak{z}_{0,n}=1$ and $\mathfrak{z}_{1,n}=(-1)^n\zeta_{2(p-1)}$ with multiplicity $\mathfrak{q}_{1,n}=p^{n-1}$.
(b) **Induction on** i **for** $2\le i\le p-1$. Assume that, for $1\le j\le i-1$, we have $\zeta_{p^n}^{(j)}=\Lambda_{j,n}$. In other words, the maximal slope $\mathfrak{s}_{j,n}$ of the Newton polygon $\mathcal{N}ewt\left(\Phi^{(j,n)}\right)$ of the jth approximation polynomial is $\frac{j}{p^{n-1}(p-1)}$ and the jth residue polynomial $\mathfrak{A}_{j,n}(T)$ has a root $\mathfrak{z}_{j,n}=\frac{(-1)^{jn}}{j!}\zeta_{2(p-1)}^{j}\in\bar{\mathbb{F}}_p$ with multiplicity $\mathfrak{q}_{j,n}=p^{n-1}$. We describe the Newton polygon of the ith approximation polynomial $\Phi^{(i,n)}(T)$ as follows.

   By the induction hypothesis and Proposition 2.7, for $1\le j\le i-1$, the Newton polygons of $\Phi^{(j,n)}(T)$ and $\Phi^{(j+1,n)}(T)$ are identical in the range $[0,p^{n-1}(p-1)-\mathfrak{q}_j]$, i.e. $[0,p^{n-1}(p-2)]$. Therefore, the Newton polygons $\mathcal{N}ewt\left(\Phi^{(i,n)}\right)$ and $\mathcal{N}ewt\left(\Phi^{(1,n)}\right)$ are identical in the range $[0,p^{n-1}(p-2)]$, and $p^{n-1}(p-2)$ is a breakpoint of the Newton polygon $\mathcal{N}ewt\left(\Phi^{(i,n)}\right)$. As a result, we only need to consider $\mathcal{N}ewt\left(\Phi^{(i,n)}\right)$ in the range

$$[p^{n-1}(p-2),p^{n-1}(p-1)].$$

In other words, we need to estimate the p-adic valuation of $b^{(i,n)}_{p^{n-1}(p-1)-k}$ for $0 \leq k \leq p^{n-1}$.

By the transfinite Newton algorithm and the assumption $\zeta^{(i-1)}_{p^n} = \Lambda_{i-1,n}$, we can obtain the formula for the coefficients $b^{(i,n)}_{p^{n-1}(p-1)-k}$ of the ith approximation polynomials $\Phi^{(i,n)}$ with $0 \leq k \leq p^{n-1}$ and their p-adic valuation can be calculated by the estimation of the p-adic valuation of $\Lambda^{p^{n-1}}_{i-1,n} - 1$ and $\Lambda^{p^n}_{i-1,n} - 1$ established in Sect. 3.3 (cf. Propositions 3.21 and 3.22, respectively), which relies on the arithmetic properties of incomplete exponential Bell polynomial studied in Sect. 3.2.

i. If $k = 0$, we have

$$b^{(i,n)}_{p^{n-1}(p-1)} = \sum_{l=0}^{p-1} \Lambda^{lp^{n-1}}_{i-1,n} = \frac{\Lambda^{p^n}_{i-1,n} - 1}{\Lambda^{p^{n-1}}_{i-1,n} - 1}.$$

By Propositions 3.21 and 3.22, we have

$$\begin{aligned} b^{(i,n)}_{p^{n-1}(p-1)} &= \frac{\frac{(-1)^{i-1}}{i!}\zeta^{i}_{2(p-1)} p^{1+\frac{i}{p-1}} + o\left(p^{1+\frac{i}{p-1}}\right)}{\sum_{l=1}^{i-1} \frac{(-1)^l}{[l!]}\zeta^{l}_{2(p-1)} p^{\frac{l}{p-1}} + O\left(p^{1+\frac{1}{p(p-1)}}\right)} \\ &= \frac{\frac{(-1)^{i-1}}{i!}\zeta^{i}_{2(p-1)} p^{1+\frac{i}{p-1}} + o\left(p^{1+\frac{i}{p-1}}\right)}{-\zeta_{2(p-1)} p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)} \\ &= \frac{(-1)^i}{i!}\zeta^{i-1}_{2(p-1)} p^{1+\frac{i-1}{p-1}} + o\left(p^{1+\frac{i-1}{p-1}}\right). \end{aligned}$$

ii. If $1 \leq k \leq p^{n-1}$, we have

$$\begin{aligned} b^{(i,n)}_{p^{n-1}(p-1)-k} &= \sum_{l=1}^{p-1} \binom{p^{n-1}l}{k} \Lambda^{p^{n-1}l-k}_{i-1,n} \\ &= \sum_{l=1}^{p-1} \left(p^{n-1}l \frac{(-1)^{k-1}}{k} + O(p^n) \right) \Lambda^{p^{n-1}l-k}_{i-1,n} \\ &= \frac{(-1)^{k-1}p^{n-1}}{k\Lambda^{k}_{i-1,n}} \sum_{l=1}^{p-1} l\Lambda^{p^{n-1}l}_{i-1,n} + O(p^n). \end{aligned}$$

Together with the elementary identity $\sum_{l=1}^{p-1} l\Lambda_{i-1,n}^{p^{n-1}l} = \frac{p\Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}}-1} - \Lambda_{i-1,n}^{p^{n-1}} \frac{\Lambda_{i-1,n}^{p^n}-1}{(\Lambda_{i-1,n}^{p^{n-1}}-1)^2}$, we obtain

$$b_{p^{n-1}(p-1)-k}^{(i,n)} = \frac{(-1)^{k-1}p^{n-1}}{k\Lambda_{i-1,n}^k} \left(\frac{p\Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}}-1} - \Lambda_{i-1,n}^{p^{n-1}} \frac{\Lambda_{i-1,n}^{p^n}-1}{(\Lambda_{i-1,n}^{p^{n-1}}-1)^2} \right) + O(p^n).$$

One can use again the estimation of the p-adic valuation of $\Lambda_{i-1,n}^{p^n} - 1$ and $\Lambda_{i-1,n}^{p^{n-1}} - 1$ in Sect. 3.3 to deduce that

$$\begin{aligned} v_p\left(b_{p^{n-1}(p-1)-k}^{(i,n)}\right) &= v_p\left(\frac{(-1)^{k-1}p^{n-1}}{k\Lambda_{i-1,n}^k} \frac{p\Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}}-1} \right) \\ &= \begin{cases} n - v_p(k) - \frac{1}{p-1} \geq 1 + \frac{i-1}{p-1}, & 1 \leq k < p^{n-1}; \\ 1 - \frac{1}{p-1}, & k = p^{n-1}, \end{cases} \end{aligned} \tag{3.2}$$

and

$$\begin{aligned} C_{\frac{p-2}{p-1}}\left(b_{p^{n-1}(p-1)-p^{n-1}}^{(i,n)}\right) &= C_{\frac{p-2}{p-1}}\left(\frac{(-1)^{p-1}p^{n-1}}{p^{n-1}\Lambda_{i-1,n}^{p^{n-1}}} \frac{p\Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}}-1} \right) \\ &= -\zeta_{2(p-1)}^{-1}. \end{aligned} \tag{3.3}$$

In conclusion, the Newton polygon of the ith approximation polynomial $\Phi^{(i,n)}$ (Fig. 2) has three breakpoints: 0, $p^{n-1}(p-2)$ and $p^{n-1}(p-1)$, with maximal breakpoint $\mathfrak{m}_{i,n} = p^{n-1}(p-2)$ and maximal slope

$$\mathfrak{s}_{i,n} = \frac{v_p\left(b_{p^{n-1}(p-1)}^{(i,n)}\right) - v_p\left(b_{p^{n-1}(p-2)}^{(i,n)}\right)}{p^{n-1}(p-1) - p^{n-1}(p-2)} = \frac{i}{p^{n-1}(p-1)}.$$

The ith residue polynomial

$$\mathfrak{A}_{i,n}(T) = -\zeta_{2(p-1)}^{-1}T^{p^{n-1}} + \frac{(-1)^i}{i!}\zeta_{2(p-1)}^{i-1}$$

has $\mathfrak{z}_{i,n} = \frac{(-1)^{in}}{i!}\zeta_{2(p-1)}^{i}$ as a root with multiplicity $\mathfrak{q}_r = p^{n-1}$.

(c) **Induction on** $i \geq p$ **for** $\zeta_{p^n}^{(i)}$. For the initial term $i = p$, the transfinite Newton algorithm and the results proved in the previous steps imply that

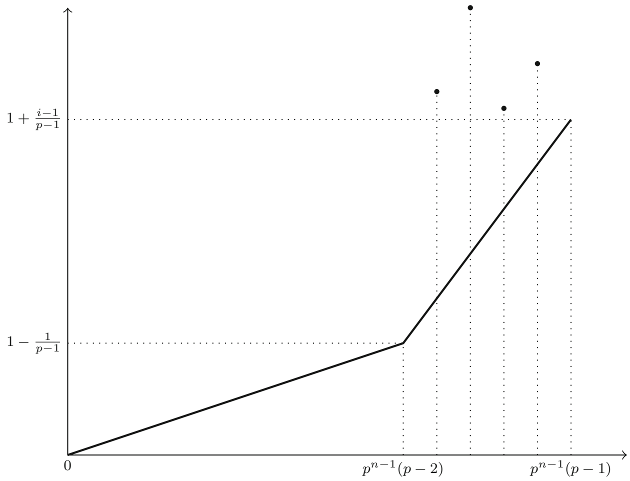

Fig. 2 $\mathscr{N}ewt\left(\Phi^{(i,n)}\right)$, $2 \le i \le p-1$

1. The Newton polygons $\mathscr{N}ewt\left(\Phi^{(p,n)}\right)$ and $\mathscr{N}ewt\left(\Phi^{(p-1,n)}\right)$ are identical in the range $[0, p^{n-1}(p-2)]$;
2. $p^{n-1}(p-2)$ is a breakpoint of $\mathscr{N}ewt\left(\Phi^{(p,n)}\right)$.

Therefore, we only need to consider $\mathscr{N}ewt\left(\Phi^{(p,n)}\right)$ in the range

$$[p^{n-1}(p-2), p^{n-1}(p-1)],$$

i.e. to estimate the p-adic valuation of $b^{(p,n)}_{p^{n-1}(p-1)-k}$ for $0 \le k \le p^{n-1}$. We express $b^{(p,n)}_{p^{n-1}(p-1)-k}$ in terms of $\Lambda_{p-1,n}$ as follows:

$$b^{(p,n)}_{p^{n-1}(p-1)-k} = \begin{cases} \sum_{l=0}^{p-1} \Lambda^{lp^{n-1}}_{p-1,n} = \frac{\Lambda^{p^n}_{p-1,n}-1}{\Lambda^{p^{n-1}}_{p-1,n}-1} & \text{if } k=0; \\ \frac{(-1)^{k-1}p^{n-1}}{k\Lambda^k_{p-1,n}} \left(\frac{p\Lambda^{p^n}_{p-1,n}}{\Lambda^{p^{n-1}}_{p-1,n}-1} - \Lambda^{p^{n-1}}_{p-1,n} \frac{\Lambda^{p^n}_{p-1,n}-1}{(\Lambda^{p^{n-1}}_{p-1,n}-1)^2} \right) + O(p^n) & \text{if } 1 \le k \le p^{n-1}. \end{cases}$$

Again using the estimation of the p-adic valuation of $\Lambda^{p^{n-1}}_{p-1,n}-1$ and $\Lambda^{p^n}_{p-1,n}-1$ in Sect. 3.3, we have

(i) $b^{(p,n)}_{p^{n-1}(p-1)} = \dfrac{\zeta_{2(p-1)}p^{2+\frac{1}{p-1}-\frac{1}{p}} + O\left(p^{2+\frac{1}{p-1}}\right)}{-\zeta_{2(p-1)}p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)} = -p^{2-\frac{1}{p}} + o\left(p^{2-\frac{1}{p}}\right),$

(ii) $$v_p\left(b^{(p,n)}_{p^{n-1}(p-1)-k}\right) = n - v_p(k) - v_p\left(\Lambda^{p^{n-1}}_{p-1,n}-1\right) = n - v_p(k) - \frac{1}{p-1} \text{ for } k = 1, \ldots, p^{n-1}-1,$$

$$\text{(iii)}\quad b^{(p,n)}_{p^{n-1}(p-2)} = (1+o(1))\frac{(-1)^{p^{n-1}-1}p}{-\zeta_{2(p-1)}p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)}$$
$$= -\zeta^{-1}_{2(p-1)}p^{\frac{p-2}{p-1}} + o\left(p^{\frac{p-2}{p-1}}\right).$$

In other words, the valuation of $b^{(p,n)}_{p^{n-1}(p-1)-k}$ is given by

$$v_p\left(b^{(p,n)}_{p^{n-1}(p-1)-k}\right) = \begin{cases} 2-\frac{1}{p} & \text{if } k=0; \\ n - v_p(k) - \frac{1}{p-1} & \text{if } 1 \le k < p^{n-1}; \\ \frac{p-2}{p-1} & \text{if } k = p^{n-1}, \end{cases}$$

the coefficient of $b^{(p,n)}_{p^{n-1}(p-1)}$ at $2-\frac{1}{p}$ is equal to -1, and the coefficient of $b^{(p,n)}_{p^{n-1}(p-2)}$ at $\frac{p-2}{p-1}$ is equal to $-\zeta^{-1}_{2(p-1)}$.
Notice that the segment $L_{p,n}$ with endpoints

$$\left(p^{n-1}(p-2), v_p\left(b^{(p,n)}_{p^{n-1}(p-2)}\right)\right) = \left(p^{n-1}(p-2), \frac{p-2}{p-1}\right)$$

and

$$\left(p^{n-1}(p-1), v_p\left(b^{(p,n)}_{p^{n-1}(p-1)}\right)\right) = \left(p^{n-1}(p-1), \frac{2p-1}{p}\right)$$

has slope

$$\frac{1}{p^{n-1}}\left(2-\frac{1}{p}-\frac{p-2}{p-1}\right) = \frac{1}{p^{n-2}(p-1)} - \frac{1}{p^n}$$

and, for all $k \in \{1, 2, \ldots, p^{n-1}-1\}$,

$$n - v_p(k) - \frac{1}{p-1} \ge \frac{p-2}{p-1} + \left(\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^n}\right)$$
$$\times \left(\left(p^{n-1}(p-1)-k\right) - p^{n-1}(p-2)\right).$$

In conclusion, $L_{p,n}$ is the segment of the Newton polygon $\mathscr{N}ewt\left(\Phi^{(p,n)}\right)$ with maximal slope $\mathfrak{s}_{p,n} = \frac{1}{p^{n-2}(p-1)} - \frac{1}{p^n}$. Therefore, we have

$$\mathfrak{A}_{p,n}(T) = -\zeta^{-1}_{2(p-1)}T^{p^{n-1}} - 1,$$

which has $\mathfrak{z}_{p,n} = (-1)^n\zeta_{2(p-1)}$ as a root with multiplicity $\mathfrak{q}_{p,n} = p^{n-1}$.

Now let $i \ge p+1$. Suppose, for all $2 \le l \le i-1$, the theorem holds; i.e. we have $\zeta^{(i-1)}_{p^n} = \Lambda_{i-1,n}$ and $\mathfrak{q}_{i-1,n} = p$. Similar to the previous case, by induction, we may assume that $\mathscr{N}ewt\left(\Phi^{(i,n)}\right)$ and $\mathscr{N}ewt\left(\Phi^{(1,n)}\right)$ are identical in the range $[0, p^{n-1}(p-2)]$. Therefore, we are reduced to consider $\mathscr{N}ewt\left(\Phi^{(i,n)}\right)$ in the range $[p^{n-1}(p-2), p^{n-2}(p-1)]$.

By the induction hypothesis $\zeta_{p^n}^{(i-1)} = \Lambda_{i-1,n}$, we get

$$b^{(i,n)}_{p^{n-1}(p-1)-k} = \begin{cases} \sum_{l=0}^{p-1} \Lambda_{i-1,n}^{lp^{n-1}} = \frac{\Lambda_{i-1,n}^{p^n}-1}{\Lambda_{i-1,n}^{p^{n-1}}-1}, & \text{if } k=0; \\ \frac{(-1)^{k-1}p^{n-1}}{k\Lambda_{i-1,n}^k}\left(\frac{p\Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}}-1} - \Lambda_{i-1,n}^{p^{n-1}}\frac{\Lambda_{i-1,n}^{p^n}-1}{(\Lambda_{i-1,n}^{p^{n-1}}-1)^2}\right) + O(p^n), & \text{if } 1 \le k \le p^{n-1}. \end{cases}$$

Again using the estimation of the p-adic valuation of $\Lambda_{i-1,n}^{p^n} - 1$ and $\Lambda_{i-1,n}^{p^{n-1}} - 1$ in Sect. 3.3, we have

i.

$$b^{(i,n)}_{p^{n-1}(p-1)} = \frac{\zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+1}}} + O\left(p^{2+\frac{1}{p-1}}\right)}{-\zeta_{2(p-1)} p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)} = -p^{2-\frac{1}{p^{i-p+1}}} + o\left(p^{2-\frac{1}{p^{i-p+1}}}\right),$$

ii.

$$v_p\left(b^{(i,n)}_{p^{n-1}(p-1)-k}\right) = n - v_p(k) - v_p\left(\Lambda_{p-1,n}^{p^{n-1}} - 1\right) = n - v_p(k) - \frac{1}{p-1}, \text{ for } k = 1, \ldots, p^{n-1}-1,$$

iii.

$$b^{(i,n)}_{p^{n-1}(p-2)} = (1+o(1))\frac{(-1)^{p^{n-1}-1}p}{-\zeta_{2(p-1)} p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)} = -\zeta_{2(p-1)}^{-1} p^{\frac{p-2}{p-1}} + o\left(p^{\frac{p-2}{p-1}}\right).$$

In other words, the valuation of $b^{(i,n)}_{p^{n-1}(p-1)-k}$ is given by

$$v_p\left(b^{(i,n)}_{p^{n-1}(p-1)-k}\right) = \begin{cases} 2 - \frac{1}{p^{i-p+1}} & \text{if } k=0; \\ n - v_p(k) - \frac{1}{p-1} & \text{if } 1 \le k < p^{n-1}; \\ \frac{p-2}{p-1} & \text{if } k = p^{n-1}, \end{cases}$$

the coefficient of $b^{(i,n)}_{p^{n-1}(p-1)}$ at $2-\frac{1}{p^{i-p+1}}$ is equal to -1, and the coefficient of $b^{(i,n)}_{p^{n-1}(p-1)}$ at $\frac{p-2}{p-1}$ is equal to $-\zeta^{-1}_{2(p-1)}$. Notice that the segment $L_{i,n}$ with endpoints

$$\left(p^{n-1}(p-2), v_p\left(b^{(i,n)}_{p^{n-1}(p-2)}\right)\right)=\left(p^{n-1}(p-2), \frac{p-2}{p-1}\right)$$

and

$$\left(p^{n-1}(p-1), v_p\left(b^{(i,n)}_{p^{n-1}(p-1)}\right)\right)=\left(p^{n-1}(p-1), 2-\frac{1}{p^{i-p+1}}\right)$$

has slope

$$\frac{1}{p^{n-1}}\left(2-\frac{1}{p^{i-p+1}}-\frac{p-2}{p-1}\right)=\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{i-p+n}}$$

and, for all $k\in\{1,2,\ldots,p^{n-1}-1\}$,

$$\begin{aligned}n-v_p(k)-\frac{1}{p-1}\geq\frac{p-2}{p-1}&+\left(\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{i-p+n}}\right)\\&\times\left(\left(p^{n-1}(p-1)-k\right)-p^{n-1}(p-2)\right).\end{aligned}$$

We conclude that $L_{i,n}$ is the segment of $\mathscr{N}\!ewt\left(\Phi^{(i,n)}\right)$ (Fig. 3) with maximal

$$\mathfrak{s}_{i,n}=\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{i-p+n}}.$$

Therefore, we have

$$\mathfrak{A}_{i,n}(T)=-\zeta^{-1}_{2(p-1)}T^{p^{n-1}}-1,$$

which has $\mathfrak{z}_{i,n}=(-1)^n\zeta_{2(p-1)}$ as a root with multiplicity $\mathfrak{q}_{i,n}=p^{n-1}$.

Proof of 2: The proof of the first assertion shows that $\aleph_0\left(\zeta_{p^n}\right)$ is completely determined once we have fixed our choice of $\mathfrak{z}_{1,n}$. As we mentioned in Definition 3.1, there are $p-1$ different candidates for $\mathfrak{z}_{1,n}$: $(-1)^n\zeta^{2k+1}_{2(p-1)}$, $k=0,1,\ldots,p-2$. Therefore, there exist $p-1$ different elements $m_0,\ldots,m_{p-2}\in(\mathbb{Z}/p^n\mathbb{Z})^\times$ such that

$$\aleph_0\left(\zeta^{m_k}_{p^n}\right)=\sum_{i=0}^{p-1}\frac{\left((-1)^n\zeta^{2k+1}_{2(p-1)}\right)^i}{[i!]}p^{\frac{i}{p^{n-1}(p-1)}}+\sum_{j=n}^{\infty}(-1)^n\zeta^{2k+1}_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^j}}.$$

If there exist two different elements m_{t_1}, m_{t_2} in $\mathcal{R}_n$ such that $m_{t_1}\equiv m_{t_2} \pmod p$, then by abuse of notations, we assume $m_{t_1}, m_{t_2}\in\mathbb{Z}$ and $m_{t_1}=ph+m_{t_2}$, where h

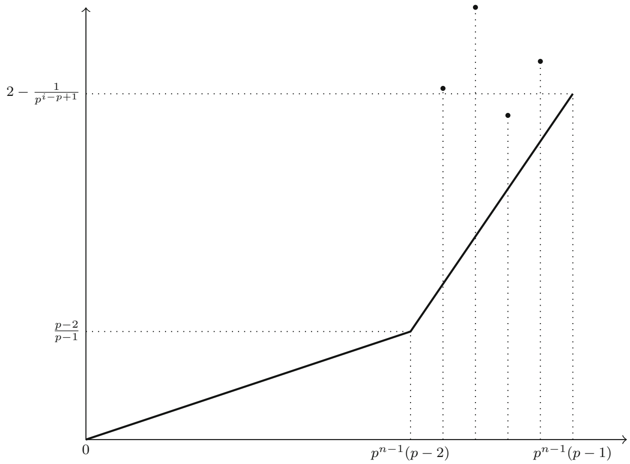

Fig. 3 $\mathcal{N}ewt\left(\Phi^{(i,n)}\right)$, $i \geq p$

is a positive integer. Then, we have $\zeta_{p^n}^{m_{t_1}} = \zeta_{p^n}^{m_{t_2}} \cdot \left(\zeta_{p^n}^{p}\right)^h$, where $\left(\zeta_{p^n}^{p}\right)^h$ is a p^{n-1}-th root of unity. Set $\left(\zeta_{p^n}^{p}\right)^h = \zeta_{p^{n-1}}^{r}$ with r a positive integer. By Proposition 3.2 and Proposition 3.2, we have $\zeta_{p^{n-1}} = 1 + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right)$ for all $n \geq 2$. Therefore,

$$\zeta_{p^{n-1}}^{r} = \left(1 + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right)\right)^r = 1 + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right),$$

and consequently

$$\zeta_{p^n}^{m_{t_1}} = \zeta_{p^n}^{m_{t_2}} \cdot \left(1 + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right)\right) = \zeta_{p^n}^{m_{t_2}} + O\left(p^{\frac{1}{p^{n-2}(p-1)}}\right),$$

i.e. $\aleph_0\left(\zeta_{p^n}^{m_{t_1}}\right) = \aleph_0\left(\zeta_{p^n}^{m_{t_2}}\right)$, which contradicts our assumption. Now the result follows from the fact that there are exactly $p-1$ mod p residue classes in $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

Proof of 3: Similar to the proof of the second assertion, we can prove that for $\tilde{m} \in (\mathbb{Z}/p^n\mathbb{Z})^\times$, $\aleph_0\left(\zeta_{p^n}^{m}\right) = \aleph_0\left(\zeta_{p^n}^{\tilde{m}}\right)$ is equivalent to $m \equiv \tilde{m} \pmod{p}$. By the second assertion, we can find such $\tilde{m}$ in $\mathcal{R}_n$. ◻

Instead of using the Newton algorithm directly, we explore the canonical expansion of ζ_p in $\mathbb{L}_p$ by using the expansion of ζ_{p^2}:

Proposition 3.4 *The canonical expansion of ζ_p is given as follows: $\zeta_p = \sum_{k=0}^{\infty}[c_k]p^{\frac{k}{p-1}}$, with $c_k \in \mathbb{F}_{p^2}$ for all $k \in \mathbb{Z}_{\geq 0}$. In particular, for $0 \leq k \leq p-1$, we have $c_k = (-1)^k \frac{\zeta_{2(p-1)}^k}{k!}$.*

Proof The first assertion, as a direct consequence of Lemma 3.5, is proved by Lampert (cf. [11]). Since $\zeta_{p^2}^p$ is a primitive pth root, we may assume $\zeta_{p^2}^p = \zeta_p^r$ for some $r \in \{1, 2, \ldots, p-1\}$. On the one hand, we calculate

$$\begin{aligned}
\left(\zeta_{p^2}\right)^p &= \left(\sum_{k=0}^{p-1} \frac{\zeta_{2(p-1)}^k}{[k!]} p^{\frac{1}{p(p-1)}} + O\left(p^{\frac{1}{p-1} - \frac{1}{p^2}} \right) \right)^p \\
&= \sum_{k=0}^{p-1} \left(\frac{\zeta_{2(p-1)}^k}{[k!]} p^{\frac{1}{p(p-1)}} \right)^p + o(p) \\
&= \sum_{k=0}^{p-1} (-1)^k \frac{\zeta_{2(p-1)}^k}{[k!]} p^{\frac{k}{p-1}} + o(p).
\end{aligned} \tag{3.4}$$

On the other hand, since $\zeta_p = 1 - \zeta_{2(p-1)} p^{\frac{1}{p-1}} + o\left(p^{\frac{1}{p-1}} \right)$ (cf. Proposition 3.2 of local cite), we have

$$\zeta_p^r = \left(1 - \zeta_{2(p-1)} p^{\frac{1}{p-1}} + o\left(p^{\frac{1}{p-1}} \right) \right)^r = 1 - [r] \zeta_{2(p-1)} p^{\frac{1}{p-1}} + o\left(p^{\frac{1}{p-1}} \right). \tag{3.5}$$

By comparing (3.4) and (3.5), we know that $r = 1$ and consequently

$$\zeta_p = \zeta_{p^2}^p = \sum_{k=0}^{p-1} \left[(-1)^k \frac{\zeta_{2(p-1)}^k}{k!} \right] p^{\frac{k}{p-1}} + o(p).$$

$\square$

Lemma 3.5 *Let $p \geq 3$ be a prime number. Then, we have $\mathbb{Q}_p\left(\zeta_p\right) = \mathbb{Q}_p\left(\zeta_{2(p-1)} p^{\frac{1}{p-1}}\right)$.*

Proof Since $\mathbb{Q}_p\left(\zeta_{2(p-1)} p^{\frac{1}{p-1}}\right)$ and $\mathbb{Q}_p\left(\zeta_p\right)$ have the same degree over $\mathbb{Q}_p$, we only need to show $\mathbb{Q}_p\left(\zeta_{2(p-1)} p^{\frac{1}{p-1}}\right) \subseteq \mathbb{Q}_p\left(\zeta_p\right)$. It is enough to show $x^{p-1} = \frac{(\zeta_p - 1)^{p-1}}{-p}$ has a solution in $\mathbb{Q}_p\left(\zeta_p\right)$.

By Proposition 3.2, we have

$$\frac{(\zeta_p - 1)^{p-1}}{-p} = -p^{-1} \left(1 - \zeta_{2(p-1)} p^{\frac{1}{p-1}} + o\left(p^{\frac{1}{p-1}} \right) - 1 \right)^{p-1} = 1 + o\left(p^0 \right);$$

thus, we may set $\frac{(\zeta_p - 1)^{p-1}}{-p} = 1 + M$, where M is in the maximal ideal of $\mathbb{Q}_p\left(\zeta_p\right)$. Since $\binom{\frac{1}{p-1}}{k} \in \mathbb{Z}_p$, the binomial series $(1 + M)^{\frac{1}{p-1}} = \sum_{k=0}^{\infty} \binom{\frac{1}{p-1}}{k} M^k$ converges in $\mathbb{Q}_p\left(\zeta_p\right)$. $\square$

Since we apply the transfinite Newton algorithm for every $n \geq 2$ independently and the first $\aleph_0$ terms of the expansion is determined by $\mathfrak{z}_{1,n}$, if we take the same $\zeta_{2(p-1)}$ for every $\mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)}$, $n = 1, 3, \ldots$, the following result should be noticed:

Corollary 3.6 *For every $n \geq 1$, we have $\aleph_0\left(\zeta_{p^n}\right) = \aleph_0\left(\zeta_{p^{n+1}}^p\right)$.*

Proof Since ζ_{p^n} and $\zeta_{p^{n+1}}^p$ are both p^n-th primitive roots of unity, by Theorem 3.3, we only need to check that

$$C_{\frac{1}{p^{n-1}(p-1)}}\left(\zeta_{p^{n+1}}^p\right) = \mathfrak{z}_{1,n} = (-1)^n \zeta_{2(p-1)}.$$

One calculates

$$\begin{aligned}
\zeta_{p^{n+1}}^p &= \left(1 + (-1)^{n-1}\zeta_{2(p-1)}p^{\frac{1}{p^n(p-1)}} + O\left((-1)^{n-1}\zeta_{2(p-1)}p^{\frac{2}{p^n(p-1)}}\right)\right)^p \\
&= 1 + \left((-1)^{n-1}\zeta_{2(p-1)}p^{\frac{1}{p^n(p-1)}}\right)^p + \left(O\left((-1)^{n-1}\zeta_{2(p-1)}p^{\frac{2}{p^n(p-1)}}\right)\right)^p + O(p) \\
&= 1 + (-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-1}(p-1)}} + O\left(p^{\frac{2}{p^{n-1}(p-1)}}\right),
\end{aligned}$$

and consequently, $C_{\frac{1}{p^{n-1}(p-1)}}\left(\zeta_{p^{n+1}}^p\right) = (-1)^n\zeta_{2(p-1)}$, as expected. □

3.2 Bell polynomials and Stirling numbers of the second kind

In this paragraph, we introduce the notion of incomplete exponential Bell polynomials and Stirling numbers of the second kind, whose arithmetic properties will be used to estimate the p-adic valuation of $\Lambda_{i,n}^{p^{n-1}} - 1$ and $\Lambda_{i,n}^{p^n} - 1$ in Sect. 3.3.

Generalities

The Bell polynomials are used to study set partitions in combinatorial mathematics. Let $\alpha_l = (j_1, j_2, \ldots, j_l) \in \mathbb{N}^l$ be a multi-index. We denote its norm by $|\alpha_l| = j_1 + j_2 + \cdots + j_l$ and its factorial by $\alpha_l! = \prod_{k=1}^l j_k!$. Let $\mathbf{x} = (x_1, \ldots, x_l)$ be a l-tuple of formal variables. The power of a multi-index α_l of $\mathbf{x}$ is defined by

$$\mathbf{x}^{\alpha_l} := \prod_{i=1}^{l} x_i^{j_i}.$$

Definition 3.7 For integer numbers $n \geq k \geq 0$, the **incomplete exponential Bell polynomial** with parameter (n, k) is a polynomial given by

$$\begin{aligned}
&B_{n,k}(x_1, x_2, \ldots, x_{n-k+1}) \\
&:= \sum_{\substack{\alpha_{n-k+1}=(j_1,\ldots,j_{n-k+1})\in\mathbb{N}^{n-k+1} \\ |\alpha_{n-k+1}|=k, \sum_{i=1}^{n-k+1} i j_i = n}} \frac{n!}{\alpha_{n-k+1}!}\left(\frac{x_1}{1!}, \ldots, \frac{x_{n-k+1}}{(n-k+1)!}\right)^{\alpha_{n-k+1}}.
\end{aligned}$$

With multinomial theorem, the incomplete exponential Bell polynomial can also be defined in terms of its generating function (cf. [5, P.134 Theorem A]):

$$\frac{1}{k!}\left(\sum_{m\geq 1} x_m \frac{t^m}{m!}\right)^k = \sum_{n\geq k} B_{n,k}(x_1,\ldots,x_{n-k+1})\frac{t^n}{n!},\ k=0,1,2,\ldots. \tag{3.6}$$

From the algebraic point of view, the Bell polynomials can be computed using its generating function. In particular, if k is small or close to n, the Bell polynomial $B_{n,k}(x_1,\ldots,x_{n-k+1})$ is easy to compute

Lemma 3.8 – $B_{n,k}(x_1,\ldots,x_{n-k+1}) = \begin{cases} x_n & \text{if } k=1; \\ \frac{1}{2}\sum_{t=1}^{n-1}\binom{n}{t}x_t x_{n-t} & \text{if } k=2. \end{cases}$

– $B_{n,k}(x_1,\ldots,x_{n-k+1}) = \begin{cases} (x_1)^n & \text{if } k=n; \\ \binom{n}{2}(x_1)^{n-2}x_2 & \text{if } k=n-1; \\ \binom{n}{3}(x_1)^{n-3}x_3 + 3\binom{n}{4}(x_1)^{n-4}(x_2)^2 & \text{if } k=n-2. \end{cases}$

The special values of the incomplete exponential Bell polynomial at the points $(1,\ldots,1)$ and $(\overbrace{1,\ldots,1}^{r},0,\ldots,0)$ are called Stirling numbers of the second kind and r-restricted Stirling numbers of the second kind (cf. [10,14]), respectively. More precisely, we have the following definition.

Definition 3.9 1. For integer numbers $n\geq k\geq 0$, the **Stirling number of the second kind** is defined by

$$\left\{ {n \atop k} \right\} = B_{n,k}(1,1,\ldots,1);$$

2. For integer numbers $n\geq k\geq 0$ and positive integer r, the r-**restricted Stirling number of the second kind** is defined by

$$\left\{ {n \atop k} \right\}_{\leq r} = \begin{cases} \left\{ {n \atop k} \right\}, & \text{if } n-k+1\leq r; \\ B_{n,k}(\overbrace{1,\ldots,1}^{r},0,\ldots,0), & \text{otherwise.} \end{cases}$$

Using the generating function formula Lemma 3.6 for Bell polynomials, one has

Lemma 3.10 (Generating function) *For $k\in\mathbb{N}$, then we have*

1. $\frac{1}{k!}\left(\sum_{m\geq 1}\frac{t^m}{m!}\right)^k = \sum_{n\geq k}\left\{ {n \atop k} \right\}\frac{t^n}{n!}$;
2. $\frac{1}{k!}\left(\sum_{m=1}^{r}\frac{t^m}{m!}\right)^k = \sum_{n=k}^{rk}\left\{ {n \atop k} \right\}_{\leq r}\frac{t^n}{n!}$.

By comparing Lemma 3.6 and the second assertion of Lemma 3.10, we have

Corollary 3.11 *If $n \geq rk+1$, then we have $\left\{ {n \atop k} \right\}_{\leq r} = 0$. Therefore, we can rewrite the second assertion of Lemma 3.10 as follows:*

$$\frac{1}{k!}\left(\sum_{m=1}^{r}\frac{t^m}{m!}\right)^k = \sum_{n=k}^{\infty}\left\{ {n \atop k} \right\}_{\leq r}\frac{t^n}{n!}, \quad k = 0, 1, 2, \ldots .$$

We denote by $(x)_n = x(x-1)(x-2)\ldots(x-n+1)$ the falling factorials, which form a basis of the $\mathbb{Q}$-vector space $\mathbb{Q}[x]$. The Stirling numbers of the second kind may also be characterized as the coordinate of powers of the indeterminate x with respect to the basis consisting of the falling factorials (cf. [5, p. 207, Theorem B]) : If $n > 0$, one has

$$x^n = \sum_{m=0}^{n}\left\{ {n \atop m} \right\}(x)_m. \tag{3.7}$$

Corollary 3.12

$$\sum_{k=1}^{n}(-1)^{k-1}(k-1)!\left\{ {n \atop k} \right\} = \begin{cases} 0, & n \geq 2; \\ 1, & n = 1. \end{cases}$$

Proof When $n = 1$, the assertion follows from direct calculation.

When $n \geq 2$, since $\binom{x}{k} = \binom{x-1}{k-1}\frac{x}{k}$ and $\left\{ {n \atop 0} \right\} = 0$, by Lemma 3.7, we know that

$$\sum_{k=1}^{n}\left\{ {n \atop k} \right\}\binom{x-1}{k-1}(k-1)! = x^{n-1}.$$

By setting $x = 0$, we have

$$\sum_{k=1}^{n}\left\{ {n \atop k} \right\}\binom{-1}{k-1}(k-1)! = 0,$$

where $\binom{-1}{k-1} = (-1)^{k-1}$. □

Arithmetic properties

Now we establish several lemmas related to the arithmetic properties of (restricted) Stirling numbers of the second kind. The first lemma (cf. Lemma 3.13) summarizes several well-known facts about the arithmetic properties of binomial coefficients, and the other lemmas (cf. Lemmas 3.14, 3.15, 3.16 and 3.17) characterize the mod p congruence properties of some special (restricted) Stirling numbers of the second kind, which will be used in Propositions 3.19, 3.20, 3.18, and 3.22.

Lemma 3.13 *Let $p \geq 3$ be a prime number and $a, b \in \mathbb{N}$ be two natural numbers such that $a \geq b$. If n is an integer satisfying $1 \leq n \leq p-1$ and k is a positive integer, then we have*

1. $v_p\left(\binom{p^n}{a}\right) = n - v_p(a)$;
2. $\binom{pk}{n} \equiv pk\frac{(-1)^{n-1}}{n} \mod p^2$;
3. $\binom{ap}{bp} \equiv \binom{a}{b} \mod p^2$.

Proof The first and the second assertions are well known. The third assertion can be found in [8, Theorem 1.6]. □

Lemma 3.14 *Let p be an odd prime number. For an integer k that $1 \leq k \leq p$, one has*

$$\left\{ \begin{matrix} p-1+k \\ p \end{matrix} \right\} \equiv \begin{cases} 1 & \text{if } k = 1 \text{ or } p; \\ 0 & \text{otherwise} \end{cases} \pmod{p}.$$

Proof By [2, Theorem 5.2], we have

$$\left\{ \begin{matrix} n \\ ap^m \end{matrix} \right\} \equiv \begin{cases} \binom{\frac{n-ap^{m-1}}{p-1}-1}{\frac{n-ap^m}{p-1}} & \text{if } n \equiv a \pmod{p-1}, \\ 0 & \text{otherwise.} \end{cases} \pmod{p^m}$$

for positive integers n, a, m that $m \geq 1$, $a > 0$ and $n \geq ap^m$. The assertion follows by taking $n = p-1+k$ and $a = m = 1$ in the above formula. □

Lemma 3.15 *Let p be an odd prime number and r an integer number satisfying $1 \leq r < p-1$, then one has*

$$\left\{ \begin{matrix} r+p \\ p \end{matrix} \right\}_{\leq r} = B_{r+p,p}(1, \ldots, 1, 0) \equiv 0 \mod p.$$

Proof If $r = 1$, then $p+1 \geq 1 \cdot p + 1$ and the result follows from Corollary 3.11.

Now we suppose $r \geq 2$. By [6, (1.3)], one has the following identity:

$$B_{n,k}(x_1, \ldots, x_{n-k+1}) = \frac{1}{x_1} \cdot \frac{1}{n-k} \sum_{\alpha=1}^{n-k} \binom{n}{\alpha} \left((k+1) - \frac{n+1}{\alpha+1}\right) x_{\alpha+1} \times B_{n-\alpha,k}(x_1, \ldots, x_{n-\alpha-k+1}).$$

Let $n = r+p$, $k = p$ and $a_t = \begin{cases} 1 & \text{if } t \leq r; \\ 0 & \text{if } t > r. \end{cases}$ Then, one has

$$\left\{ \begin{matrix} r+p \\ p \end{matrix} \right\}_{\leq r} = B_{r+p,p}(a_1, \ldots, a_{r+1})$$

$$
\begin{aligned}
&= \frac{1}{r}\sum_{\alpha=1}^{r}\binom{r+p}{\alpha}\left((p+1)-\frac{r+p+1}{\alpha+1}\right)x_{\alpha+1} \\
&\quad \times B_{r+p-\alpha,p}(a_1,\ldots,a_{r-\alpha+1}) \\
&= \frac{1}{r}\sum_{\alpha=1}^{r-1}\binom{r+p}{\alpha}\left((p+1)-\frac{r+p+1}{\alpha+1}\right)\left\{ {r+p-\alpha \atop p} \right\}.
\end{aligned}
$$

Since $\alpha+1 \le r < p-1$ and $1 < r-\alpha+1 < p-1$, by Lemma 3.14, we have

$$
\left\{ {r+p-\alpha \atop p} \right\} \equiv 0 \pmod{p}.
$$

As a consequence, we have

$$
\left\{ {r+p \atop p} \right\}_{\le r} \equiv 0 \pmod{p}.
$$

□

Lemma 3.16 *Let i be an integer that $1 \le i \le p-1$ and $k \in \mathbb{Z}_{>0}$. Then for any integer $l \ge k$, we have*

$$
v_p\left(\frac{k!}{l!}\left\{ {l \atop k} \right\}_{\le i}\right) \ge 0.
$$

Proof For $j \in \mathbb{N}_{>0}$, we set $\delta_j = \begin{cases} 1 & \text{if } j \le i; \\ 0 & \text{otherwise.} \end{cases}$ Recall that the incomplete exponential Bell polynomial is defined as follows:

$$
\begin{aligned}
&B_{l,k}(x_1,x_2,\ldots,x_{l-k+1}) \\
&\quad = \sum_{\substack{\alpha_{l-k+1}=(j_1,\ldots,j_{l-k+1})\in\mathbb{N}^{l-k+1} \\ |\alpha_{l-k+1}|=k,\ \sum_{t=1}^{l-k+1} t j_t = l}} \frac{l!}{\alpha_{l-k+1}!}\left(\frac{x_1}{1!},\cdots,\frac{x_{l-k+1}}{(l-k+1)!}\right)^{\alpha_{l-k+1}},
\end{aligned}
$$

and the i-restricted Stirling numbers of the second kind $\left\{ {l \atop k} \right\}_{\le i}$ is the special value of $B_{l,k}$ at the point $\delta = (\delta_j)_{1\le j\le l-k+1}$. For $\alpha = (j_1,\ldots,j_{l-k+1}) \in \mathbb{N}^{l-k+1}$, we set

$$
F_{l,k,i}(\alpha) = \binom{k}{j_1,\ldots,j_{l-k+1}}\left(\frac{\delta_1}{1!},\ldots,\frac{\delta_{l-k+1}}{(l-k+1)!}\right)^{\alpha}.
$$

Then we have

$$\frac{k!}{l!}\left\{ \begin{matrix} l \\ k \end{matrix} \right\}_{\leq i} = \sum_{\substack{\alpha=(j_1,\dots,j_{l-k+1})\in\mathbb{N}^{l-k+1} \\ |\alpha|=k, \sum_{t=1}^{l-k+1} tj_t=l}} F_{l,k,i}(\alpha),$$

and it is enough to prove $v_p\left(F_{l,k,i}(\alpha)\right) \geq 0$ for all α in the above formula, which follows from the following discussions on the range of i:

Case 1: Suppose $l-k+1 \leq i < p$. We have $v_p\left(\frac{\delta_m}{m!}\right) = v_p(\delta_m) = 0$ for all $1 \leq m \leq l-k+1$. Therefore,

$$\begin{aligned} v_p\left(F_{l,k,i}(\alpha)\right) &= v_p\left(\binom{k}{j_1,\dots,j_{l-k+1}}\right) - \sum_{m=1}^{l-k+1} j_m v_p(m!) \\ &= v_p\left(\binom{k}{j_1,\dots,j_{l-k+1}}\right) \geq 0. \end{aligned}$$

Case 2: Suppose $i < l-k+1$. For $\alpha = (j_1,\dots,j_{l-k+1}) \in \mathbb{N}^{l-k+1}$, if there exists m such that $i < m \leq l-k+1$ and $j_m > 0$, then $F_{l,k,i}(\alpha) = 0$. If $j_{i+1} = \dots = j_{l-k+1} = 0$, then

$$\begin{aligned} v_p\left(F_{l,k,i}(\alpha)\right) &= v_p\left(\binom{k}{j_1,\dots,j_{l-k+1}}\right) - \sum_{m=1}^{i} j_m v_p(m!) \\ &= v_p\left(\binom{k}{j_1,\dots,j_{l-k+1}}\right) \geq 0. \end{aligned}$$

□

Lemma 3.17 *For $n \in \mathbb{N}_{\geq 2}$, $1 \leq s \leq p-1$ and $sp^{n-2} \leq t \leq p^{n-1}-1$, we have*

$$\frac{(sp^{n-2})!}{t!}\left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1} \equiv \begin{cases} 0 \mod p & \text{if } p^{n-2} \nmid t \\ \frac{s!}{(t/p^{n-2})!}\left\{ \begin{matrix} t/p^{n-2} \\ s \end{matrix} \right\} \mod p & \text{if } p^{n-2} \mid t. \end{cases}$$

Proof When $n = 2$, the assertion follows from the fact $t-s+1 \leq p-1$ and $\left\{ \begin{matrix} t \\ s \end{matrix} \right\}_{\leq p-1} = \left\{ \begin{matrix} t \\ s \end{matrix} \right\}$.

Suppose $n \geq 3$. For $1 \leq s \leq p-1$ and $sp^{n-2} \leq t$, we set $u_{s,t} = \min\{t - sp^{n-2} + 1, p-1\}$. By the definition of restricted Stirling number of the second kind, we have

$$\frac{(sp^{n-2})!}{t!}\left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1} = \sum_{\substack{\alpha=(j_1,\dots,j_{u_{s,t}})\in\mathbb{N}^{u_{s,t}} \\ |\alpha|=sp^{n-2}, \sum_{m=1}^{u_{s,t}} mj_m=t}} \binom{sp^{n-2}}{j_1,\dots,j_{u_{s,t}}}\left(\frac{1}{1!},\dots,\frac{1}{u_{s,t}!}\right)^{\alpha}.$$

By separating this sum into two parts, we can write

$$\begin{aligned}\frac{(sp^{n-2})!}{t!}\left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\le p-1} &= \sum_{\substack{\alpha=(j_1,\ldots,j_{u_{s,t}})\in (p^{n-2}\mathbb{N})^{u_{s,t}} \\ |\alpha|=sp^{n-2}, \sum_{m=1}^{u_{s,t}} mj_m=t}} \binom{sp^{n-2}}{j_1,\ldots,j_{u_{s,t}}}\left(\frac{1}{1!},\ldots,\frac{1}{u_{s,t}!}\right)^{\alpha} \\ &\quad + \sum_{\substack{\alpha=(j_1,\ldots,j_{u_{s,t}})\in \mathbb{N}^{u_{s,t}}\setminus(p^{n-2}\mathbb{N})^{u_{s,t}} \\ |\alpha|=sp^{n-2}, \sum_{m=1}^{u_{s,t}} mj_m=t}} \binom{sp^{n-2}}{j_1,\ldots,j_{u_{s,t}}} \\ &\quad \times \left(\frac{1}{1!},\ldots,\frac{1}{u_{s,t}!}\right)^{\alpha}.\end{aligned}$$

If $\alpha = (j_1,\ldots,j_{u_{s,t}}) \in \mathbb{N}^{u_{s,t}}\setminus\left(p^{n-2}\mathbb{N}\right)^{u_{s,t}}$, then, since $\binom{sp^{n-2}}{j_m}$ is a factor of $\binom{sp^{n-2}}{j_1,\ldots,j_{u_{s,t}}}$ for all $1\le m\le u_{s,t}$, and $\binom{sp^{n-2}}{j_m}$ is divided by p if $p^{n-2}\nmid j_m$, we have $\binom{sp^{n-2}}{j_1,\ldots,j_{u_{s,t}}}\left(\frac{1}{1!},\ldots,\frac{1}{u_{s,t}!}\right)^{\alpha}$ is divisible by p. Therefore, we have

$$\begin{aligned}\frac{(sp^{n-2})!}{t!}\left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\le p-1} &= \sum_{\substack{\alpha=(j_1,\ldots,j_{u_{s,t}})\in (p^{n-2}\mathbb{N})^{u_{s,t}} \\ |\alpha|=sp^{n-2}, \sum_{m=1}^{u_{s,t}} mj_m=t}} \binom{sp^{n-2}}{j_1,\ldots,j_{u_{s,t}}} \\ &\quad \times \left(\frac{1}{1!},\ldots,\frac{1}{u_{s,t}!}\right)^{\alpha} + O(p).\end{aligned} \tag{3.8}$$

By replacing j_m with $\widehat{j_m} = j_m/p^{n-2}$ and replacing α with $\widehat{\alpha} = \left(\widehat{j_1},\ldots,\widehat{j_{u_{s,t}}}\right)$, we can rewrite (3.8) as follows:

$$\begin{aligned}&\frac{(sp^{n-2})!}{t!}\left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\le p-1} \\ &= \sum_{\substack{\widehat{\alpha}=(\widehat{j_1},\ldots,\widehat{j_{u_{s,t}}})\in \mathbb{N}^{u_{s,t}} \\ |\widehat{\alpha}|=s, \sum_{m=1}^{u_{s,t}} m\widehat{j_m}=t/p^{n-2}}} \binom{sp^{n-2}}{\widehat{j_1}p^{n-2},\ldots,\widehat{j_{u_{s,t}}}p^{n-2}} \\ &\quad \times \left(\left(\frac{1}{1!}\right)^{p^{n-2}},\ldots,\left(\frac{1}{u_{s,t}!}\right)^{p^{n-2}}\right)^{\widehat{\alpha}} + O(p).\end{aligned} \tag{3.9}$$

Notice that we have the identity

$$\begin{aligned}\binom{sp^{n-2}}{\widehat{j_1}p^{n-2},\ldots,\widehat{j_{u_{s,t}}}p^{n-2}} &= \binom{\widehat{j_1}p^{n-2}+\cdots+\widehat{j_{u_{s,t}}}p^{n-2}}{\widehat{j_1}p^{n-2}} \\ &\quad \times \binom{\widehat{j_2}p^{n-2}+\cdots+\widehat{j_{u_{s,t}}}p^{n-2}}{\widehat{j_2}p^{n-2}}\cdots\binom{\widehat{j_{u_{s,t}}}p^{n-2}}{\widehat{j_{u_{s,t}}}p^{n-2}},\end{aligned}$$

and by applying the formula $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^2}$ (cf. Lemma 3.13) to this identity, we obtain

$$\begin{aligned}\binom{sp^{n-2}}{\widehat{j}_1 p^{n-2}, \dots, \widehat{j}_{u_{s,t}} p^{n-2}} &= \binom{\widehat{j}_1 + \cdots + \widehat{j}_{u_{s,t}}}{\widehat{j}_1}\binom{\widehat{j}_2 + \cdots + \widehat{j}_{u_{s,t}}}{\widehat{j}_2} \cdots \binom{\widehat{j}_{u_{s,t}}}{\widehat{j}_{u_{s,t}}} \\ &\quad + O(p) \\ &= \binom{s}{\widehat{j}_1, \cdots, \widehat{j}_{u_{s,t}}} + O(p).\end{aligned}$$

In addition, for all $m \in \{1, \dots, u_{s,t}\}$, we have

$$\left(\frac{1}{m!}\right)^{\widehat{j}_m p^{n-2}} = \left(\frac{1}{m!}\right)^{\widehat{j}_m} + O(p).$$

Therefore, we can rewrite (3.9) as follows:

$$\begin{aligned}\frac{(sp^{n-2})!}{t!}\left\{ t \atop sp^{n-2} \right\}_{\leq p-1} = \sum_{\substack{\widehat{\alpha}=(\widehat{j}_1,\dots,\widehat{j}_{u_{s,t}})\in\mathbb{N}^{u_{s,t}} \\ |\widehat{\alpha}|=s, \sum_{m=1}^{u_{s,t}} m\widehat{j}_m = t/p^{n-2}}} \binom{s}{\widehat{j}_1, \dots, \widehat{j}_{u_{s,t}}}\left(\frac{1}{1!}, \cdots, \frac{1}{u_{s,t}!}\right)^{\widehat{\alpha}} \\ + O(p). \qquad (3.10)\end{aligned}$$

If $p^{n-2} \nmid t$, then the summation above is void, and consequently,

$$v_p\left(\frac{(sp^{n-2})!}{t!}\left\{ t \atop sp^{n-2} \right\}_{\leq p-1}\right) \geq 1.$$

It remains to deal with the case $p^{n-2} \mid t$. By setting $t = \widehat{t} p^{n-2}$ with $s \leq \widehat{t} \leq p-1$, we have

$$\frac{(sp^{n-2})!}{t!}\left\{ t \atop sp^{n-2} \right\}_{\leq p-1} = \frac{(sp^{n-2})!}{(\widehat{t}p^{n-2})!}\left\{ \widehat{t}p^{n-2} \atop sp^{n-2} \right\}_{\leq p-1}.$$

We conclude our assertion in this case by the following discussion on the relation between $\widehat{t}$ and s.

1. If $\widehat{t} = s$, we have

$$\frac{(sp^{n-2})!}{(\widehat{t}p^{n-2})!}\left\{ \widehat{t}p^{n-2} \atop sp^{n-2} \right\}_{\leq p-1} = 1 = \frac{s!}{\widehat{t}!}\left\{ \widehat{t} \atop s \right\}.$$

2. If $\widehat{t} > s$, we have $\widehat{t}p^{n-2} - sp^{n-2} + 1 \geq p^{n-2} + 1 > p - 1$. Therefore, $u_{s,\widehat{t}p^{n-2}} = p - 1$ and

$$\frac{(sp^{n-2})!}{(\widehat{t}p^{n-2})!}\left\{\begin{matrix}\widehat{t}p^{n-2}\\ sp^{n-2}\end{matrix}\right\}_{\leq p-1} = \sum_{\substack{\widehat{\alpha}=(\widehat{j}_1,\ldots,\widehat{j}_{p-1})\in\mathbb{N}^{p-1}\\ |\widehat{\alpha}|=s,\sum_{m=1}^{p-1} m\widehat{j}_m=\widehat{t}}} \binom{s}{\widehat{j}_1,\ldots,\widehat{j}_{p-1}} \times \left(\frac{1}{1!},\ldots,\frac{1}{(p-1)!}\right)^{\widehat{\alpha}} + O(p).$$

If there exists $p-1 \geq r > \widehat{t} - s + 1$ such that $j_r \neq 0$, then

$$1\widehat{j}_1 + \cdots + (p-1)\widehat{j}_{p-1} \geq 1 \cdot (s-1) + r > \widehat{t},$$

which contradicts to the condition that $\sum_{m=1}^{p-1} m\widehat{j}_m = \widehat{t}$. Therefore, $\widehat{j}_r = 0$ for all $r > \widehat{t} - s + 1$. As a consequence,

$$\begin{aligned}
&\frac{(sp^{n-2})!}{(\widehat{t}p^{n-2})!}\left\{\begin{matrix}\widehat{t}p^{n-2}\\ sp^{n-2}\end{matrix}\right\}_{\leq p-1}\\
&= \sum_{\substack{\widehat{\alpha}=(\widehat{j}_1,\ldots,\widehat{j}_{\widehat{t}-s+1})\in\mathbb{N}^{\widehat{t}-s+1}\\ |\widehat{\alpha}|=s,\sum_{m=1}^{\widehat{t}-s+1} m\widehat{j}_m=\widehat{t}}} \binom{s}{\widehat{j}_1,\ldots,\widehat{j}_{\widehat{t}-s+1}} \left(\frac{1}{1!},\ldots,\frac{1}{(\widehat{t}-s+1)!}\right)^{\widehat{\alpha}} + O(p)\\
&= \frac{s!}{\widehat{t}!}\left\{\begin{matrix}\widehat{t}\\ s\end{matrix}\right\} + O(p).
\end{aligned}$$

□

The main technical propositions

In this paragraph, we establish our main technical propositions (cf. Proposition 3.18, Proposition 3.19 and Proposition 3.20) using the arithmetic properties of (restricted) Stirling numbers of the second kind.

Proposition 3.18 *For $n \in \mathbb{N}_{\geq 2}$, we have*

$$\left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!}\zeta_{2(p-1)}^{l} p^{\frac{l}{p^{n-1}(p-1)}}\right)^{p^{n-1}} - 1 = \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]}\zeta_{2(p-1)}^{l} p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p(p-1)}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

Proof Let $\lambda_n = (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-1}(p-1)}}$, and we rewrite left-hand side of the equality as follows:

$$\left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!} \zeta_{2(p-1)}^{l} p^{\frac{l}{p^{n-1}(p-1)}} \right)^{p^{n-1}} - 1 = \left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!} \right)^{p^{n-1}} + H(n), \quad (3.11)$$

where $H(n) = \sum_{j=1}^{p^{n-1}-1} \binom{p^{n-1}}{j} \left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!} \right)^j$.

Note that $v_p\left(\binom{p^{n-1}}{j} \left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!} \right)^j \right) = n-1-v_p(j)+\frac{j}{p^{n-1}(p-1)}$ and the condition

$$n - 1 - v_p(j) + \frac{j}{p^{n-1}(p-1)} < 1 + \frac{1}{p-1}$$

implies $v_p(j) = n-2$. We can rewrite $H(n)$ as follows:

$$\sum_{s=1}^{p-1} \binom{p^{n-1}}{sp^{n-2}} \left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!} \right)^{sp^{n-2}} + O\left(p^{1+\frac{1}{p-1}}\right). \quad (3.12)$$

Using Lemma 3.13, one can further simplify it as follows:

$$H(n) = \sum_{s=1}^{p-1} \binom{p}{s} \left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!} \right)^{sp^{n-2}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

Applying the generating function formula for restricted Stirling numbers of the second kind, we obtain

$$\begin{aligned} H(n) &= \sum_{s=1}^{p-1} \binom{p}{s} (sp^{n-2})! \sum_{t=sp^{n-2}}^{\infty} \left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1} \frac{\lambda_n^t}{t!} + O\left(p^{1+\frac{1}{p-1}}\right) \\ &= \sum_{s=1}^{p-1} \binom{p}{s} \sum_{t=sp^{n-2}}^{\infty} \left(\frac{(sp^{n-2})!}{t!} \left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1} \right) \lambda_n^t + O\left(p^{1+\frac{1}{p-1}}\right). \end{aligned} \quad (3.13)$$

By Lemma 3.16, we know that $\frac{(sp^{n-2})!}{t!} \left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1}$ has non-negative valuation. Note that we have $v_p(\lambda_n) = \frac{1}{p^{n-1}(p-1)}$, and for $t \geq p^{n-1}$, we have $v_p\left(\lambda_n^t\right) \geq \frac{1}{p-1}$. Thus, we can assemble the terms with $t \geq p^{n-1}$ of $H(n)$ into the error term:

$$H(n) = \sum_{s=1}^{p-1} \binom{p}{s} \sum_{t=sp^{n-2}}^{p^{n-1}-1} \left(\frac{(sp^{n-2})!}{t!} \left\{ \begin{matrix} t \\ sp^{n-2} \end{matrix} \right\}_{\leq p-1} \right) \lambda_n^t + O\left(p^{1+\frac{1}{p-1}}\right). \quad (3.14)$$

We denote by $\widehat{t} = \frac{t}{p^{n-2}}$. By Lemma 3.17, we obtain

$$H(n) = \sum_{s=1}^{p-1} \binom{p}{s} \sum_{\widehat{t}=s}^{p-1} \left(\frac{s!}{\widehat{t}!} \left\{ \begin{matrix} \widehat{t} \\ s \end{matrix} \right\} \right) \lambda_n^{\widehat{t} p^{n-2}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

By exchanging the order of the summations and using the second assertion of Lemma 3.13, we have

$$\begin{aligned} H(n) =& p \sum_{\widehat{t}=1}^{p-1} \frac{\lambda_n^{\widehat{t} p^{n-2}}}{\widehat{t}!} \sum_{s=1}^{\widehat{t}} (-1)^{s-1}(s-1)! \left\{ \begin{matrix} \widehat{t} \\ s \end{matrix} \right\} + O\left(p^{1+\frac{1}{p-1}}\right) \\ =& p\lambda_n^{p^{n-2}} + O\left(p^{1+\frac{1}{p-1}}\right), \end{aligned} \tag{3.15}$$

where the last equality follows from Corollary 3.12.

For the term $\left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!}\right)^{p^{n-1}}$, by multinomial theorem, one has

$$\left(\sum_{l=1}^{p-1} \frac{\lambda_n^l}{l!}\right)^{p^{n-1}} = \sum_{\substack{j_1,\ldots,j_{p-1}\in\mathbb{N} \\ j_1+\cdots+j_{p-1}=p^{n-1}}} \binom{p^{n-1}}{j_1,\ldots,j_{p-1}} \prod_{l=1}^{p-1} \left(\frac{\lambda_n^l}{l!}\right)^{j_l}.$$

If $j_1, \ldots, j_{p-1} < p^{n-1}$, then we have

$$\begin{aligned} v_p\left(\binom{p^{n-1}}{j_1,\ldots,j_{p-1}} \prod_{l=1}^{p-1} \left(\frac{\lambda_n^l}{l!}\right)^{j_l}\right) &= v_p\left(\binom{p^{n-1}}{j_1,\ldots,j_{p-1}}\right) + \sum_{l=1}^{p-1} l j_l v_p(\lambda_n) \\ &\geq 1 + \frac{1}{p^{n-1}(p-1)} \sum_{l=1}^{p-1} 1 \cdot j_l \\ &= 1 + \frac{1}{p-1}. \end{aligned}$$

If there exists a $l \in \{1, \ldots, p-1\}$ such that $j_l = p^{n-1}$, one calculates

$$\begin{aligned} \left(\frac{\lambda_n^l}{l!}\right)^{p^{n-1}} &= (-1)^l \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} \frac{1}{(l!)^{p^{n-1}}} = (-1)^l \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} \left(\frac{1}{[l!]} + O(p)\right) \\ &= \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + O\left(p^{1+\frac{1}{p-1}}\right). \end{aligned}$$

In conclusion, we have

$$\begin{aligned}\left(\sum_{l=1}^{p-1}\frac{\lambda_n^l}{l!}\right)^{p^{n-1}} &= \sum_{l=1}^{p-1}\left(\frac{\lambda_n^l}{l!}\right)^{p^{n-1}} + O\left(p^{1+\frac{1}{p-1}}\right)\\ &= \sum_{l=1}^{p-1}\frac{(-1)^l}{[l!]}\zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + O\left(p^{1+\frac{1}{p-1}}\right).\end{aligned}$$

Combining with (3.11) and (3.15), we have

$$\begin{aligned}&\left(\sum_{l=0}^{p-1}\frac{(-1)^{ln}}{l!}\zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}}\right)^{p^{n-1}} - 1\\ &= \left(\sum_{l=1}^{p-1}\frac{\lambda_n^l}{l!}\right)^{p^{n-1}} + H(n)\\ &= \sum_{l=1}^{p-1}\frac{(-1)^l}{[l!]}\zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p(p-1)}} + O\left(p^{1+\frac{1}{p-1}}\right),\end{aligned}$$

as expected. □

Proposition 3.19 *For $n \in \mathbb{N}_{\geq 2}$, we have*

$$\left(\sum_{l=0}^{p-1}\frac{(-1)^l}{l!}\zeta_{2(p-1)}^l p^{\frac{l}{p-1}}\right)^p - 1 = O\left(p^{2+\frac{1}{p-1}}\right).$$

Proof Let $\theta_n = -\zeta_{2(p-1)} p^{\frac{1}{p-1}}$, then by the generate function of the restricted Stirling number of the second kind, we have

$$\left(\sum_{l=0}^{p-1}\frac{\theta_n^l}{l!}\right)^p - 1 = \sum_{j=1}^{p}\binom{p}{j}\left(\sum_{l=1}^{p-1}\frac{\theta_n^l}{l!}\right)^j = \sum_{j=1}^{p}\binom{p}{j}\sum_{k=j}^{\infty}\frac{j!}{k!}\left\{\begin{matrix}k\\ j\end{matrix}\right\}_{\leq p-1}\theta_n^k. \tag{3.16}$$

Notice that

$$v_p\left(\binom{p}{j}\left(\frac{j!}{k!}\left\{\begin{matrix}k\\ j\end{matrix}\right\}_{\leq p-1}\right)\theta_n^k\right) \geq 1 - v_p(j) + \frac{k}{p-1},$$

by assembling terms with valuation equal or greater than $2+\frac{1}{p-1}$, we can rewrite (3.16) as follows:

$$\left(\sum_{l=0}^{p-1}\frac{\theta_n^l}{l!}\right)^p-1=\sum_{j=1}^{p-1}\binom{p}{j}\sum_{k=j}^{p-1}\frac{j!}{k!}\left\{ {k \atop j} \right\}_{\le p-1}\theta_n^k+\sum_{k=p}^{2p-2}\frac{p!}{k!}\left\{ {k \atop p} \right\}_{\le p-1}\theta_n^k$$
$$+\,O\left(p^{2+\frac{1}{p-1}}\right).$$

By the definition of restricted Stirling number of the second kind and changing the order of summations, we can further reduce this to

$$\begin{aligned}\left(\sum_{l=0}^{p-1}\frac{\theta_n^l}{l!}\right)^p-1&=\sum_{k=1}^{p-1}\frac{\theta_n^k}{k!}\sum_{j=1}^{k}\binom{p}{j}j!\left\{ {k \atop j} \right\}+\sum_{k=p}^{2p-2}\frac{p!}{k!}\left\{ {k \atop p} \right\}\theta_n^k+O\left(p^{2+\frac{1}{p-1}}\right)\\&=p\sum_{k=1}^{p-1}\frac{\theta_n^k}{k!}\sum_{j=1}^{k}(-1)^{j-1}(j-1)!\left\{ {k \atop j} \right\}+\sum_{k=p}^{2p-2}\frac{p!}{k!}\left\{ {k \atop p} \right\}\theta_n^k\\&\quad+\,O\left(p^{2+\frac{1}{p-1}}\right).\end{aligned}$$

By Corollary 3.12,

$$p\sum_{k=1}^{p-1}\frac{\theta_n^k}{k!}\sum_{j=1}^{k}(-1)^{j-1}(j-1)!\left\{ {k \atop j} \right\}=p\theta_n.$$

On the other hand, since $v_p(p!)=v_p(k!)=1$ for $k=p,\ldots 2p-2$, by Lemma 3.14, we have

$$\frac{p!}{k!}\left\{ {k \atop p} \right\}=\begin{cases}O(p) & \text{if } p<k\le 2p-2;\\ 1+O(p) & \text{if } k=p,\end{cases}$$

and consequently,

$$\left(\sum_{l=0}^{p-1}\frac{\theta_n^l}{l!}\right)^p-1=p\theta_n+\theta_n^p+O\left(p^{2+\frac{1}{p-1}}\right)=O\left(p^{2+\frac{1}{p-1}}\right).$$

$\square$

Proposition 3.20 *Let p be a prime and let $1\le i<p-1$ be an integer. For $1\le l\le i+1$ an integer, we set*

$$G_i(l)=\left(\sum_{k=1}^{l}(-1)^{k-1}(k-1)!\left\{ {l \atop k} \right\}_{\le i}\right)+\frac{p\cdot l!}{(l+p-1)!}\left\{ {l+p-1 \atop p} \right\}_{\le i}.$$

Then, we have

$$G_i(l) = \begin{cases} -1 + O(p) & \textit{if } l = i + 1; \\ O(p) & \textit{if } l \leq i. \end{cases}$$

Proof We rewrite $G_i(l)$ as follows:

$$G_i(l) = \begin{cases} \sum_{k=1}^{l}(-1)^{k-1}(k-1)!\left\{ {l \atop k} \right\} + \left\{ {p-1+l \atop p} \right\}\frac{p\cdot l!}{(l+p-1)!} & \text{if } l \leq i; \\ \sum_{k=1}^{i+1}(-1)^{k-1}(k-1)!\left\{ {i+1 \atop k} \right\}_{\leq i} + \frac{p\cdot(i+1)!}{(i+p)!}\left\{ {i+p \atop p} \right\}_{\leq i} & \text{if } l = i+1. \end{cases}$$

Recall that, the Corollary 3.12 says

$$\sum_{k=1}^{n}(-1)^{k-1}(k-1)!\left\{ {n \atop k} \right\} = \begin{cases} 0, & n \geq 2; \\ 1, & n = 1. \end{cases}$$

- Suppose $l \leq i$. If $l = 1$, then one has

$$G_i(1) = 1 + \left\{ {p \atop p} \right\}\frac{1}{(p-1)!} \equiv 0 \mod p.$$

  If $1 < l \leq i < p-1$, by Lemma 3.14 and Corollary 3.12, one has

$$G_i(n) = 0 + \left\{ {p-1+l \atop p} \right\}\frac{p\cdot l!}{(l+p-1)!} \equiv 0 \mod p.$$

- Suppose $l = i + 1$, by Lemma 3.15 and Corollary 3.12, one has

$$G_i(i+1) = \sum_{k=1}^{i+1}(-1)^{k-1}(k-1)!\left\{ {i+1 \atop k} \right\}_{\leq i} + \frac{p\cdot(i+1)!}{(i+p)!}\left\{ {i+p \atop p} \right\}_{\leq i}.$$

  For $2 \leq k \leq i+1$, one has

$$\left\{ {i+1 \atop k} \right\}_{\leq i} = \left\{ {i+1 \atop k} \right\},$$

  therefore,

$$G_i(i+1) = (-1)^{1-1}(1-1)!\left\{ {i+1 \atop 1} \right\}_{\leq i} + \sum_{k=2}^{i+1}(-1)^{k-1}(k-1)!\left\{ {i+1 \atop k} \right\}$$

$$
\begin{aligned}
&+ \left\{ \begin{matrix} i+p \\ p \end{matrix} \right\}_{\leq i} \frac{p \cdot (i+1)!}{(i+p)!} \\
&= 0 - (-1)^{1-1}(1-1)! \left\{ \begin{matrix} i+1 \\ 1 \end{matrix} \right\} + \sum_{k=1}^{i+1} (-1)^{k-1}(k-1)! \left\{ \begin{matrix} i+1 \\ k \end{matrix} \right\} \\
&+ O(p) \frac{p(i+1)!}{(i+p)!} \\
&= -1 + 0 + O(p) \\
&= -1 + O(p).
\end{aligned}
$$

□

3.3 Estimation of $\Lambda_{i,n}^{p^{n-1}} - 1$ and $\Lambda_{i,n}^{p^n} - 1$

Let $n \in \mathbb{N}_{\geq 2}$. Recall that we set

$$
\Lambda_{i,n} = \begin{cases} \sum_{k=0}^{i} \frac{(-1)^{kn}}{[k!]} \zeta_{2(p-1)}^{k} p^{\frac{k}{p^{n-1}(p-1)}} & \text{for } 0 \leq i \leq p-1, \\ \Lambda_{p-1,n} + \sum_{l=n}^{i-p+n} (-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^l}} & \text{for } i \geq p. \end{cases}
$$

As indicated in Sect. 3.1, for $i \in \mathbb{N}_{>0}$ and $0 \leq k \leq p^{n-1}$, we can describe the coefficients $b_{p^{n-1}(p-1)-k}^{(i,n)}$ of the i-approximation polynomial $\Phi^{(i,n)}$ by the following formula:

$$
\begin{aligned}
&b_{p^{n-1}(p-1)-k}^{(i,n)} \\
&= \begin{cases} \dfrac{\Lambda_{i-1,n}^{p^n} - 1}{\Lambda_{i-1,n}^{p^{n-1}} - 1} & \text{if } k = 0; \\ \dfrac{(-1)^{k-1} p^{n-1}}{k \Lambda_{i-1,n}^{k}} \left(\dfrac{p \Lambda_{i-1,n}^{p^n}}{\Lambda_{i-1,n}^{p^{n-1}} - 1} - \Lambda_{i-1,n}^{p^{n-1}} \dfrac{\Lambda_{i-1,n}^{p^n} - 1}{(\Lambda_{i-1,n}^{p^{n-1}} - 1)^2} \right) + O(p^n) & \text{if } 1 \leq k \leq p^{n-1}. \end{cases}
\end{aligned}
$$

This leads us to estimate the p-adic valuation of $\Lambda_{i,n}^{p^{n-1}} - 1$ and $\Lambda_{i,n}^{p^n} - 1$ in Propositions 3.21 and 3.22, respectively. In general, we obtain the estimation by induction, but since the formula for $\Lambda_{i,n}$ in the ranges $1 \leq i < p-1$ and $p-1 \leq i$ is different, the statements will be separated into two parts.

Proposition 3.21 *Let $n \in \mathbb{N}_{\geq 2}$.*

1. *If $1 \leq i < p-1$, we have*

$$
\Lambda_{i,n}^{p^{n-1}} - 1 = \sum_{l=1}^{i} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^{l} p^{\frac{l}{p-1}} + O\left(p^{1+\frac{1}{p(p-1)}}\right).
$$

2. *If $p-1 \leq i$, we have*

$$\Lambda_{i,n}^{p^{n-1}}-1=\sum_{l=1}^{p-1}\frac{(-1)^l}{[l!]}\zeta_{2(p-1)}^l p^{\frac{l}{p-1}}+\zeta_{2(p-1)}p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}}+O\left(p^{1+\frac{1}{p-1}}\right).$$

Proof We prove this lemma by induction on i.

1. If $i=1$, then we have

$$\begin{aligned}\Lambda_{1,n}^{p^{n-1}}-1&=\left(1+(-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-1}(p-1)}}\right)^{p^{n-1}}-1\\&=\sum_{k=1}^{p^{n-1}}\binom{p^{n-1}}{k}(-1)^{kn}\zeta_{2(p-1)}^k p^{\frac{k}{p^{n-1}(p-1)}}\\&=-\zeta_{2(p-1)}p^{\frac{1}{p-1}}+O\left(p^{1+\frac{1}{p(p-1)}}\right).\end{aligned}$$

Suppose the lemma is true for j with $1\leq j\leq i-1\leq p-3$. Then, we have

$$\begin{aligned}\Lambda_{i,n}^{p^{n-1}}-1&=\left(\Lambda_{i-1,n}+\frac{(-1)^{in}}{[i!]}\zeta_{2(p-1)}^i p^{\frac{i}{p^{n-1}(p-1)}}\right)^{p^{n-1}}-1\\&=\Lambda_{i-1,n}^{p^{n-1}}-1+\sum_{k=1}^{p^{n-1}-1}\binom{p^{n-1}}{k}\Lambda_{i-1,n}^{p^{n-1}-k}\frac{(-1)^{ikn}}{[i!]^k}\zeta_{2(p-1)}^{ik}p^{\frac{ik}{p^{n-1}(p-1)}}\\&\quad+\frac{(-1)^{inp^{n-1}}}{[i!]^{p^{n-1}}}\zeta_{2(p-1)}^{ip^{n-1}}p^{\frac{i}{p-1}}\\&=\Lambda_{i-1,n}^{p^{n-1}}-1+\frac{(-1)^i}{[i!]}\zeta_{2(p-1)}^i p^{\frac{i}{p-1}}+O\left(p^{1+\frac{i}{p-1}}\right).\end{aligned}$$

Therefore, the induction hypothesis allows us to conclude this case.

2. If $i=p-1$, then we have

$$\begin{aligned}\Lambda_{p-1,n}^{p^{n-1}}-1&=\left(\sum_{l=0}^{p-1}\frac{(-1)^{ln}}{[l!]}\zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}}\right)^{p^{n-1}}-1\\&=\left(\sum_{l=0}^{p-1}\frac{(-1)^{ln}}{l!}\zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}}+O\left(p^{1+\frac{2}{p^{n-1}(p-1)}}\right)\right)^{p^{n-1}}-1 \qquad (3.17)\\&=\left(\sum_{j=0}^{p^{n-1}}\binom{p^{n-1}}{j}\left(\sum_{l=0}^{p-1}\frac{(-1)^{ln}}{l!}\zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}}\right)^{p^{n-1}-j}\right.\end{aligned}$$

$$\times \left(O\left(p^{1+\frac{2}{p^{n-1}(p-1)}} \right) \right)^j \right) - 1.$$

For $1 \leq j \leq p^{n-1}$, we observe that

$$\begin{aligned} &\binom{p^{n-1}}{j} \left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}} \right)^{p^{n-1}-j} \left(O\left(p^{1+\frac{2}{p^{n-1}(p-1)}} \right) \right)^j \\ &\quad = O\left(p^{n-1-v_p(j)+j+\frac{2j}{p^{n-1}(p-1)}} \right). \end{aligned}$$

Since $v_p(j) \leq n-1$ and $j \geq 1$, we know that

$$n - 1 - v_p(j) + j + \frac{2j}{p^{n-1}(p-1)} > 2,$$

and thus, (3.17) can be written as follows:

$$\begin{aligned} \Lambda_{p-1,n}^{p^{n-1}} - 1 &= \left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}} \right)^{p^{n-1}} - 1 + \sum_{j=1}^{p^{n-1}} O\left(p^2\right) \\ &= \left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}} \right)^{p^{n-1}} - 1 + O\left(p^2\right). \end{aligned}$$

By Proposition 3.18, we have

$$\begin{aligned} &\left(\sum_{l=0}^{p-1} \frac{(-1)^{ln}}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p^{n-1}(p-1)}} \right)^{p^{n-1}} - 1 \\ &\quad = \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p(p-1)}} + O\left(p^{1+\frac{1}{p-1}} \right). \end{aligned}$$

As a consequence, we obtain

$$\Lambda_{p-1,n}^{p^{n-1}} - 1 = \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p}} + O\left(p^{1+\frac{1}{p-1}} \right).$$

3. Now we suppose that the formula holds for all j with $p-1 \leq j \leq i-1$, i.e.

$$\Lambda_{j,n}^{p^{n-1}} - 1 = \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{j-p+2}}} + O\left(p^{1+\frac{1}{p-1}} \right).$$

One has

$$\begin{aligned}\Lambda_{i,n}^{p^{n-1}}-1&=\left(\Lambda_{i-1,n}+(-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^{p^{n-1}}-1\\&=\Lambda_{i-1,n}^{p^{n-1}}-1+\left((-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^{p^{n-1}}\\&\quad+\sum_{k=1}^{p^{n-1}-1}\binom{p^{n-1}}{k}\Lambda_{i-1,n}^{p^{n-1}-k}\left((-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^k.\end{aligned}\tag{3.18}$$

Notice that for every $k\in\{1,\dots,p^{n-1}-1\}$,

$$\begin{aligned}&v_p\left(\binom{p^{n-1}}{k}\Lambda_{i-1,n}^{p^{n-1}-k}\left((-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^k\right)\\&\qquad=(n-1)-v_p(k)+\frac{k}{p^{n-2}}\left(\frac{1}{p-1}-\frac{1}{p^{i-p+2}}\right).\end{aligned}$$

Thus, the condition with variable k

$$v_p\left(\binom{p^{n-1}}{k}\Lambda_{i-1,n}^{p^{n-1}-k}\left((-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^k\right)<1+\frac{1}{p-1}$$

implies $k=p^{n-2}$. Since $\Lambda_{i-1,n}=1+O\left(p^{\frac{1}{p^{n-1}(p-1)}}\right)$, we have

$$\begin{aligned}&\binom{p^{n-1}}{p^{n-2}}\left((-1)^n\zeta_{2(p-1)}p^{\frac{1}{p^{n-2}(p-1)}-\frac{1}{p^{n-p+i}}}\right)^{p^{n-2}}\Lambda_{i-1,n}^{p^{n-1}-p^{n-2}}\\&=p\left((-1)^{n-2}(-1)^n\zeta_{2(p-1)}p^{\frac{1}{p-1}-\frac{1}{p^{2-p+i}}}\right)\left(1+O\left(p^{\frac{1}{p^{n-1}(p-1)}}\right)\right)^{p^{n-2}(p-1)}\\&\quad+O\left(p^2\right)\\&=\zeta_{2(p-1)}p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}}\left(1+\sum_{r=1}^{p-1}\binom{p-1}{r}O\left(p^{\frac{r}{p^{n-1}(p-1)}}\right)\right)^{p^{n-2}}+O\left(p^2\right)\\&=\zeta_{2(p-1)}p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}}\left(1+O\left(p^{\frac{1}{p^{n-1}(p-1)}}\right)\right)^{p^{n-2}}+O\left(p^2\right).\end{aligned}\tag{3.19}$$

Notice that

$$\left(1+O\left(p^{\frac{1}{p^{n-1}(p-1)}}\right)\right)^{p^{n-2}}=1+\sum_{r=1}^{p^{n-2}}\binom{p^{n-2}}{r}O\left(p^{\frac{r}{p^{n-1}(p-1)}}\right)$$

$$= 1 + \sum_{r=1}^{p^{n-2}} O\left(p^{n-2-v_p(r)+\frac{r}{p^{n-1}(p-1)}}\right)$$
$$= 1 + O\left(p^{\frac{1}{p(p-1)}}\right).$$

Since $1 + \frac{1}{p-1} - \frac{1}{p^{2-p+i}} + \frac{1}{p(p-1)} > 1 + \frac{1}{p-1}$ for all $i \geq p$, we can rewrite (3.19) as follows:

$$\zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} \left(1 + O\left(p^{\frac{1}{p(p-1)}}\right)\right) + O\left(p^2\right)$$
$$= \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

Thus, by assembling the terms of valuation $\geq 1 + \frac{1}{p-1}$ in (3.18), we obtain

$$\Lambda_{i,n}^{p^{n-1}} - 1 = \Lambda_{i-1,n}^{p^{n-1}} - 1 + \left((-1)^n \zeta_{2(p-1)} p^{\frac{1}{p^{n-2}(p-1)} - \frac{1}{p^{n-p+i}}}\right)^{p^{n-1}}$$
$$+ \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} + O\left(p^{1+\frac{1}{p-1}}\right)$$
$$= \Lambda_{i-1,n}^{p^{n-1}} - 1 - \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{1-p+i}}}$$
$$+ \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

Finally, combining with the induction hypothesis, we obtain

$$\Lambda_{i,n}^{p^{n-1}} - 1 = \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+1}}} + O\left(p^{1+\frac{1}{p-1}}\right)$$
$$- \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{1-p+i}}} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} + O\left(p^{1+\frac{1}{p-1}}\right)$$
$$= \sum_{l=1}^{p-1} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{2-p+i}}} + O\left(p^{1+\frac{1}{p-1}}\right).$$

□

Proposition 3.22 *Let $n \in \mathbb{N}_{\geq 2}$.*

1. *For $1 \leq i < p - 1$, we have*

$$\Lambda_{i,n}^{p^n} - 1 = \frac{(-1)^i}{(i+1)!} \zeta_{2(p-1)}^{i+1} p^{1+\frac{i+1}{p-1}} + o\left(p^{1+\frac{i+1}{p-1}}\right).$$

2. *For $i \geq p-1$, we have*

$$\Lambda_{i,n}^{p^n} - 1 = \zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{2+\frac{1}{p-1}}\right).$$

Proof of 1: Recall that by Proposition 3.21, for $1 \leq i < p-1$, we have

$$\Lambda_{i,n}^{p^{n-1}} = \sum_{l=0}^{i} \frac{(-1)^l}{[l!]} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + O\left(p^{1+\frac{1}{p(p-1)}}\right).$$

Let $\tilde{\Lambda}_{i,n} = \sum_{l=0}^{i} \frac{(-1)^l}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} = \sum_{l=0}^{i} \frac{\theta_n^l}{l!}$, with $\theta_n = -\zeta_{2(p-1)} p^{\frac{1}{p-1}}$. By 2 of Lemma 3.10, for $1 \leq k \leq p$, we have

$$\left(\tilde{\Lambda}_{i,n} - 1\right)^k = \left(\sum_{l=1}^{i} \frac{\theta_n^l}{l!}\right)^k = \sum_{l=k}^{ik} \frac{k!}{l!} \left\{ \begin{matrix} l \\ k \end{matrix} \right\}_{\leq i} \theta_n^l.$$

We remark that $v_p(\theta_n) = \frac{1}{p-1}$, $v_p(\tilde{\Lambda}_{i,n}) = 0$ and

$$\tilde{\Lambda}_{i,n} - \Lambda_{i,n}^{p^{n-1}} = \sum_{l=0}^{i} (-1)^l \left(\frac{1}{l!} - \frac{1}{[l!]}\right) \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} + O(p^{1+\frac{1}{p(p-1)}}).$$

For all $0 \leq l \leq i < p-1$, we have $v_p(l! - [l!]) \geq 1$; thus, we have

$$\tilde{\Lambda}_{i,n} - \Lambda_{i,n}^{p^{n-1}} = \sum_{l=1}^{i} O\left(p^{1+\frac{l}{p-1}}\right) + O\left(p^{1+\frac{1}{p(p-1)}}\right),$$

and we can rewrite $\Lambda_{i,n}^{p^n} - 1$ as follows:

$$\begin{aligned}\Lambda_{i,n}^{p^n} - 1 &= \left(\Lambda_{i,n}^{p^{n-1}}\right)^p - 1 = \left(\tilde{\Lambda}_{i,n} + O\left(p^{1+\frac{1}{p(p-1)}}\right)\right)^p - 1 \\ &= \tilde{\Lambda}_{i,n}^p - 1 + O\left(p^{2+\frac{1}{p(p-1)}}\right).\end{aligned}$$

We reduce to estimate the p-adic valuation of $\tilde{\Lambda}_{i,n}^p - 1$. On the other hand, we have

$$\tilde{\Lambda}_{i,n}^p - 1 = \sum_{k=1}^{p} \binom{p}{k} (\tilde{\Lambda}_{i,n} - 1)^k = \sum_{k=1}^{p} \binom{p}{k} \sum_{l=k}^{ik} \frac{k!}{l!} \left\{ \begin{matrix} l \\ k \end{matrix} \right\}_{\leq i} \theta_n^l. \tag{3.20}$$

By Lemma 3.16, we have $v_p\left(\frac{k!}{l!}\left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i}\right) \geq 0$ for any $k, l \in \mathbb{N}$. Thus, we can rewrite (3.20) by assembling the terms with valuation $> 1 + \frac{i+1}{p-1}$:

$$\begin{aligned}
\tilde{\Lambda}_{i,n}^{p} - 1 &= o(p^{1+\frac{i+1}{p-1}}) + \sum_{l=1}^{i+1} \frac{\theta_n^l}{l!} \sum_{k=1}^{l} \binom{p}{k} k! \left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i} + \sum_{l=p}^{p+i} \frac{p!}{l!} \left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i} \theta_n^l \\
&= o\left(p^{1+\frac{i+1}{p-1}}\right) + p \sum_{l=1}^{i+1} \frac{\theta_n^l}{l!} \sum_{k=1}^{l} (-1)^{k-1}(k-1)! \left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i} \\
&\quad - p \sum_{l=1}^{i+1} \frac{p!}{(l+p-1)!} \left\{\begin{matrix} l+p-1 \\ p \end{matrix}\right\}_{\leq i} \theta_n^l \\
&= o\left(p^{1+\frac{i+1}{p-1}}\right) + p \sum_{l=1}^{i+1} \frac{\theta_n^l}{l!} \left(\sum_{k=1}^{l} (-1)^{k-1}(k-1)! \left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i}\right. \\
&\quad \left. + \frac{l!p}{(l+p-1)!} \left\{\begin{matrix} l+p-1 \\ p \end{matrix}\right\}_{\leq i}\right),
\end{aligned} \tag{3.21}$$

where the last equality follows from $-(p-1)! \equiv 1 \mod p$. Let

$$G_i(l) = \left(\sum_{k=1}^{l} (-1)^{k-1}(k-1)! \left\{\begin{matrix} l \\ k \end{matrix}\right\}_{\leq i}\right) + \frac{p \cdot l!}{(l+p-1)!} \left\{\begin{matrix} l+p-1 \\ p \end{matrix}\right\}_{\leq i}.$$

Together with Proposition 3.20, we have

$$\begin{aligned}
\widetilde{\Lambda}_{i,n}^{p} - 1 &= o\left(p^{1+\frac{i+1}{p-1}}\right) + p \sum_{l=1}^{i+1} G_i(l) \frac{\theta_n^l}{l!} \\
&= o\left(p^{1+\frac{i+1}{p-1}}\right) + p \left(\sum_{l=1}^{i} O(p) \frac{\theta_n^l}{l!} + (-1 + O(p)) \frac{\theta_n^{i+1}}{(i+1)!}\right) \\
&= o\left(p^{1+\frac{i+1}{p-1}}\right) + p \left(o(p) - \frac{\theta_n^{i+1}}{(i+1)!} + O\left(p^{1+\frac{i+1}{p-1}}\right)\right) \\
&= o\left(p^{1+\frac{i+1}{p-1}}\right) - p \frac{\theta_n^{i+1}}{(i+1)!} \\
&= \frac{(-1)^i}{(i+1)!} \zeta_{2(p-1)}^{i+1} p^{1+\frac{i+1}{p-1}} + o\left(p^{1+\frac{i+1}{p-1}}\right).
\end{aligned}$$

As a consequence, we have

$$\Lambda_{i,n}^{p^n} - 1 = \frac{(-1)^i}{(i+1)!} \zeta_{2(p-1)}^{i+1} p^{1+\frac{i+1}{p-1}} + o\left(p^{1+\frac{i+1}{p-1}}\right).$$

Proof of 2: Now suppose $i \geq p-1$.
Let $\tilde{\Lambda}_{p-1,n} = \sum_{l=0}^{p-1} \frac{(-1)^l}{l!} \zeta_{2(p-1)}^l p^{\frac{l}{p-1}} = \sum_{l=0}^{p-1} \frac{\theta_n^l}{l!}$, with $\theta_n = -\zeta_{2(p-1)} p^{\frac{1}{p-1}}$. By Proposition 3.21, we have

$$\begin{aligned}\Lambda_{i,n}^{p^{n-1}} - \ & \tilde{\Lambda}_{p-1,n} \\ &= \sum_{l=0}^{p-1} (-1)^l \zeta_{2(p-1)}^l \left(\frac{1}{[l!]} - \frac{1}{l!} \right) p^{\frac{l}{p-1}} \\ &\quad + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{1+\frac{1}{p-1}}\right) \\ &= \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{1+\frac{1}{p-1}}\right).\end{aligned}$$

Therefore, we have

$$\begin{aligned}\Lambda_{i,n}^{p^n} - 1 &= \left(\Lambda_{i,n}^{p^{n-1}}\right)^p - 1 \\ &= \left(\tilde{\Lambda}_{p-1,n} + \zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{1+\frac{1}{p-1}}\right)\right)^p - 1 \\ &= \tilde{\Lambda}_{p-1,n}^p - 1 + \sum_{k=1}^{p} \binom{p}{k} \tilde{\Lambda}_{p-1,n}^{p-k} \left(\zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}}\right. \\ &\quad \left. + O\left(p^{1+\frac{1}{p-1}}\right)\right)^k \\ &= \tilde{\Lambda}_{p-1}^p - 1 + \tilde{\Lambda}_{p-1,n}^{p-1} \left(\zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{2+\frac{1}{p-1}}\right)\right) \\ &\quad + \sum_{k=2}^{p} \binom{p}{k} \tilde{\Lambda}_{p-1,n}^{p-k} \left(\zeta_{2(p-1)} p^{1+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{1+\frac{1}{p-1}}\right)\right)^k \\ &= \tilde{\Lambda}_{p-1,n}^p - 1 + \tilde{\Lambda}_{p-1,n}^{p-1} \zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{2+\frac{1}{p-1}}\right).\end{aligned} \tag{3.22}$$

Since $\tilde{\Lambda}_{p-1,n}^{p-1} = 1 + O\left(p^{\frac{1}{p-1}}\right)$, we may simplify (3.22) as

$$\Lambda_{i,n}^{p^n} - 1 = \tilde{\Lambda}_{p-1,n}^p - 1 + \zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{2+\frac{1}{p-1}}\right).$$

By Proposition 3.19, we have $\tilde{\Lambda}_{p-1,n}^p - 1 = \left(\sum_{l=0}^{p-1} \frac{\theta_n^l}{l!}\right)^p - 1 = O\left(p^{2+\frac{1}{p-1}}\right)$; therefore,

$$\Lambda_{i,n}^{p^n} - 1 = \zeta_{2(p-1)} p^{2+\frac{1}{p-1}-\frac{1}{p^{i-p+2}}} + O\left(p^{2+\frac{1}{p-1}}\right),$$

as expected. ◻

3.4 Uniformizer of $K_{2,n}$

In this section, we use the expansion of ζ_{p^2} to get a uniformizer of $K_{2,n}$:

Theorem 3.23

1. *The element* $\pi_{2,1} = \left(p^{\frac{1}{p}}\right)^{-1}\left(\zeta_{p^2} - \sum_{k=0}^{p-1}\frac{1}{[k!]}\zeta_{2(p-1)}^k p^{\frac{k}{p(p-1)}}\right)$ *is a uniformizer of* $K_{2,1}$.
2. *For* $m \geq 2$, *the element*

$$\pi_{2,m} = \left(p^{\frac{1}{p^m}}\right)^{-\frac{p^m-1}{p-1}}\left(\zeta_{p^2} - \sum_{k=0}^{p-1}\frac{1}{[k!]}\zeta_{2(p-1)}^k p^{\frac{k}{p(p-1)}} - \sum_{l=2}^{m}\zeta_{2(p-1)}p^{\frac{1}{p-1}-\frac{1}{p^l}}\right)$$

is a uniformizer of $K_{2,m}$.

Proof By Theorem 3.3, we know that

$$\zeta_{p^2} = \sum_{k=0}^{p-1}\frac{1}{[k!]}\zeta_{2(p-1)}^k p^{\frac{k}{p(p-1)}} + \sum_{k=2}^{\infty}\zeta_{2(p-1)}p^{\frac{1}{p-1}-\frac{1}{p^k}} + O\left(p^{\frac{1}{p-1}}\right).$$

Therefore,

$$\begin{aligned} v_p(\pi_{2,1}) &= v_p\left(\sum_{k=2}^{\infty}\zeta_{2(p-1)}p^{\frac{1}{p-1}-\frac{1}{p^k}} + O\left(p^{\frac{1}{p-1}}\right)\right) - \frac{1}{p} \\ &= \frac{1}{p-1} - \frac{1}{p^2} - \frac{1}{p} = \frac{1}{p^2(p-1)} = e_{K_{2,1}/\mathbb{Q}_p}^{-1} \end{aligned}$$

and similarly, $v_p(\pi_{2,m}) = \frac{1}{p^{m+1}(p-1)} = e_{K_{2,m}/\mathbb{Q}_p}^{-1}$ for $m \geq 2$.

To see that $\pi_{2,1} \in K_{2,1}$, we can write $\frac{1}{[k!]}\zeta_{2(p-1)}^k p^{\frac{k}{p(p-1)}}$ as $\left(\frac{1}{[k!]}p^{-\frac{k}{p}}\right)\left(\zeta_{2(p-1)}p^{\frac{1}{p-1}}\right)^k$, and consequently, $\pi_{2,1} \in \mathbb{Q}_p\left(\zeta_{p^2}, \zeta_{2(p-1)}p^{\frac{1}{p-1}}\right)$. By Lemma 3.5, this field is exactly $K_{2,1}$. Similarly, we have $\pi_{2,m} \in K_{2,m}$ for all $m \geq 2$, which finishes the proof. □

When doing the calculation, one should always make sure that the choices of ζ_{p^2} and $\zeta_{2(p-1)}$ are compatible, i.e. $\zeta_{p^2} = 1 + \zeta_{2(p-1)}p^{\frac{1}{p(p-1)}} + o\left(p^{\frac{1}{p(p-1)}}\right)$. To get over this inconvenience, one can replace $\zeta_{2(p-1)}p^{\frac{1}{p-1}}$ with $1-\zeta_p$ and use the fact that $\zeta_p = 1 - \zeta_{2(p-1)}p^{\frac{1}{p-1}} + O\left(p^{\frac{2}{p-1}}\right)$ to eliminate the appearance of $\zeta_{2(p-1)}p^{\frac{1}{p(p-1)}}$ and $\zeta_{2(p-1)}p^{\frac{1}{p-1}}$ in $\pi_{2,m}$:

Corollary 3.24

1. *The element* $\tilde{\pi}_{2,1} = \left(p^{\frac{1}{p}}\right)^{-1}\left(\zeta_{p^2} - \sum_{k=0}^{p-2}\frac{1}{[k!]}\left(\frac{1-\zeta_p}{p^{\frac{1}{p}}}\right)^k - p^{\frac{1}{p}}\right)$ *is a uniformizer of* $K_{2,1}$.

2. *For $m \geq 2$, the element*

$$\tilde{\pi}_{2,m} = \left(p^{\frac{1}{p^m}}\right)^{-\frac{p^m-1}{p-1}} \left(\zeta_{p^2} - \sum_{k=0}^{p-2} \frac{1}{[k!]} \left(\frac{1-\zeta_p}{p^{\frac{1}{p}}}\right)^k - p^{\frac{1}{p}} - (1-\zeta_p) \sum_{l=2}^{m} p^{-\frac{1}{p^l}}\right)$$

is a uniformizer of $K_{2,m}$.

Another method proposed by Lampert without proof (cf. [13]) to construct a uniformizer of $K_{2,2}$ (which can be generalized to arbitrary $K_{2,m}$ easily) is to consider the following sequence[7]:

$$\begin{cases} z_1 := \zeta_{p^2} - 1 - p^{\frac{1}{p}}, \\ z_2 := z_1^{p-1} + p^{\frac{1}{p}} - p^{\frac{2p-1}{p^2}}, \\ z_{n+1} := z_n^{p-1} - \left(\left[C_{v_p(z_n)}(z_n)\right] p^{v_p(z_n)}\right)^{p-1} \text{ for } n = 2, 3, \ldots. \end{cases}$$

Then, we can prove by keeping track of Supp (z_n) that

Proposition 3.25 1. *There exists an integer $N \leq p$ such that $p^3(p-1)v_p(z_N)$ is an integer satisfying*

$$p^3(p-1)v_p(z_N) \equiv -p+1 \pmod{p^2}.$$

2. *Let $M = p(p-1)(p-2)\sum_{i=1}^{N-1} v_p(z_n) + p$. Then $M \in \mathbb{Z}$ and for any solution $(a, b, c) \in \mathbb{Z}^3$ of the linear equation*

$$\left(p^2M + 1 - p\right)a + p(p-1)b + p^2c = 1,$$

the element $z_N^a \cdot p^{b/p^2} \cdot \left(\zeta_{p^2} - 1\right)^c$ is a uniformizer of $K_{2,2}$. In particular, one may take $(a, b, c) = (p+1, -pM, -2M+1)$ and

$$\pi'_{2,2} = \frac{z_N^{p+1}}{p^{M/p}\left(\zeta_{p^2} - 1\right)^{2M-1}}$$

is a uniformizer of $K_{2,2}$.

Example 3.26 When $p = 7$, our method (cf. Theorem 3.23 and Corollary 3.24) gives two uniformizers of $K_{2,2}$:

$$\pi_{2,2} = 7^{-\frac{8}{49}} \left(\zeta_{49} - 1 - \zeta_{12}7^{\frac{1}{42}} - \frac{1}{[2]}\zeta_6 7^{\frac{1}{21}} + \zeta_4 7^{\frac{1}{14}} - \frac{1}{[3]}\zeta_3 7^{\frac{2}{21}} - \zeta_{12}^5 7^{\frac{5}{42}} \right.$$
$$\left. - 7^{\frac{1}{7}} - \zeta_{12} 7^{\frac{43}{294}}\right),$$

[7] We modify Lampert's original idea slightly to correct and simplify the result.

$$\tilde{\pi}_{2,2} = 7^{-\frac{8}{49}} \left(\zeta_{49} - 1 - \left(\frac{1-\zeta_7}{7^{\frac{1}{7}}} \right) - \frac{1}{[2]} \left(\frac{1-\zeta_7}{7^{\frac{1}{7}}} \right)^2 + \left(\frac{1-\zeta_7}{7^{\frac{1}{7}}} \right)^3 \right.$$
$$\left. - \frac{1}{[3]} \left(\frac{1-\zeta_7}{7^{\frac{1}{7}}} \right)^4 - \left(\frac{1-\zeta_7}{7^{\frac{1}{7}}} \right)^5 - 7^{\frac{1}{7}} - (1-\zeta_7) 7^{\frac{1}{49}} \right),$$

while Lampert's method (cf. Proposition 3.25) provides a more complicated uniformizer of the same field:

$$\pi'_{2,2} = \frac{(((((((\zeta_{49} - 1 - 7^{1/7})^6 + 7^{1/7} - 7^{13/49})^6 + 7)^6 + 7^{43/7})^6 + 7^{37})^6 + 7^{1555/7})^6 + 7^{1333})^8}{7^{55987/7} (\zeta_{49} - 1)^{111973}}.$$

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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